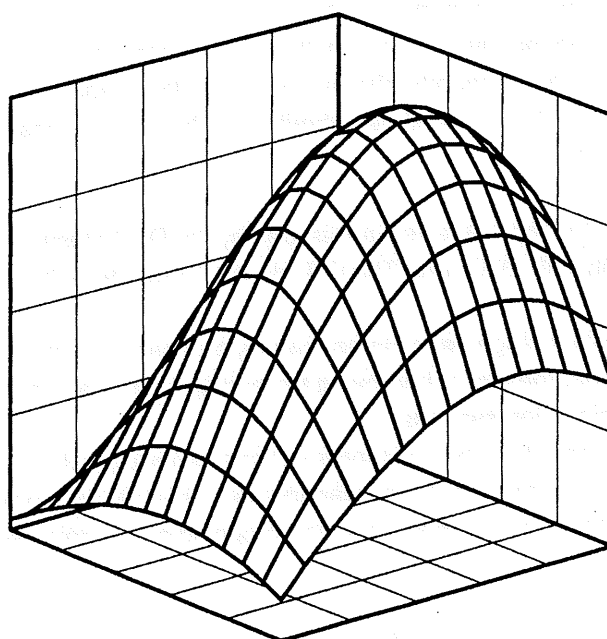


Description, parameterization and user guide of LINTUL-BRASNAP 1.1

A crop growth model of winter oilseed rape (*Brassica napus* L.)



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Description, parameterization and user guide of LINTUL- BRASNAP 1.1

A crop growth model of winter oilseed rape
(*Brassica napus* L.)

B. Habekotté

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B. Habekotté

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Samenvatting

Dit rapport beschrijft het model LINTUL-BRASNAP, dat de groei en ontwikkeling simuleert van winterkoolzaad (*Brassica napus* L.) onder potentiële groeiomstandigheden met optimale water- en nutriëntenvoorziening en zonder groeibeperkende invloed van ziekten, plagen en onkruiden. Het model is ontwikkeld binnen een studie gericht op de kwantificering van de opbrengstvorming van winterkoolzaad en op het identificeren van gewaseigenschappen die een positieve invloed hebben op de zaadopbrengst in West Europa.

Het rapport beschrijft de structuur van het model en in het kort de evaluatie van het model. Evaluatie en toepassing van het model worden uitgebreid behandeld in wetenschappelijke artikelen.

Kwantificering van parameters is in detail beschreven, behalve voor de parameters die in de literatuur worden onderbouwd. Een overzicht is gegeven van alle parameterwaarden. Het model is geschreven in FORTRAN-77 en het FSE-systeem (FORTRAN Simulation Environment 2.1) voor gewassimulatie vormt de simulatie-omgeving. Het FSE-systeem bestaat uit een hoofdprogramma, een systeem voor inlezen van weersgegevens en verscheidene hulpprogramma's voor het uitvoeren van specifieke taken. Het FSE-systeem kan zonder wijzigingen op zeer uiteenlopende computers worden gebruikt. De subroutine SUBNAP bevat de wetenschappelijke vergelijkingen. Acronymen en symbolen worden uitgelegd. Het rapport bevat tevens een 'listing' van het model en van bijbehorende input- en output-files.

Summary

This report contains a description of the model LINTUL-BRASNAP, which simulates growth and development of winter oilseed rape (*Brassica napus* L.) under potential growth conditions with ample supply of water and nutrients in a pest, disease and weed-free environment. The model has been developed within a study focused on quantifying the seed yield formation of winter oilseed rape (*Brassica napus* L.) and on identifying crop traits and combinations of traits for increasing seed yield potential in West Europe. The report describes the model structure and briefly gives a model evaluation. Evaluation and application of the model are described in detail in scientific papers. Quantification of model parameters is given in detail as far as not available in other publications. An overview is given of all parameter values. The model is written in FORTRAN-77, using the FSE system (FORTRAN Simulation Environment 2.1) for crop simulation. The FSE system consists of a main program, a system for weather data input and utilities for specific tasks. The FSE program can easily be run to in different computer environments. The subroutine SUBNAP contains the scientific equations. Acronyms and symbols are explained. The report also contains the listing of the model and of the input and output files.

1 Introduction

1.1 Objective and outline of the report

LINTUL-BRASNAP simulates growth and development of winter oilseed rape (*Brassica napus* L.) under potential growth conditions with ample supply of water and nutrients in a pest, disease and weed-free environment. The model has been developed within a study focused on quantifying the seed yield formation of winter oilseed rape and on identifying crop traits and combinations of traits for increasing seed yield potential in West Europe, combining experimental research with crop growth simulation (Habekotté, 1996e). Experimental research was carried out to answer questions about key processes of crop growth and seed yield formation such as light absorption, light use efficiency for total dry matter production, pod and seed set, seed filling and formation and relocation of reserve carbohydrates (Habekotté and Smid, 1992). These processes were quantified in relation to crop physiological and phenological characteristics and the environmental factors temperature and radiation (Habekotté, 1993; this report). A simulation model was used to quantify crop phenological development in relation to temperature and photoperiod (Habekotté, 1996a). The identified key processes of crop growth and phenological development were integrated in the crop growth model LINTUL-BRASNAP to explain seed yield variation under potential growth conditions in West Europe (Habekotté, 1996b). The model recognizably reproduced the observed variation in various crop characteristics, such as total dry matter production, pod density, seed density and seed yield resulting from different sowing dates and growing seasons (Habekotté, 1996b). The model was used to identify weak and strong yield-determining components of winter oilseed rape compared to winter wheat as high-yielding reference crop (Habekotté, 1996c) and for evaluating various crop traits and combinations of traits for increasing seed yield (Habekotté, 1996d).

The objective of this report is to give a detailed documentation of the model LINTUL-BRASNAP so that it can easily be used by others. Chapter 2 describes model development and model structure and briefly describes model evaluation. Chapter 3 describes quantification of model parameters as far as not described in detail in other publications. Chapter 4 gives an introduction to the FORTRAN Simulation Environment (FSE), guidelines for using the model and an overview of the source code and input and output files of the model. Parameter values are given in appendix I and acronyms and symbols are explained in appendix II. Appendix III contains the listing of the model and of the input and output files.

1.2 Winter oilseed rape

1.2.1 Distribution and products

The terms 'oilseed rape' or 'rapeseed' refer to two *Brassica* species grown for oil, *Brassica napus* L. and *Brassica campestris* L. Oilseed rape is a crop of temperate climates. There are spring-sown and autumn-sown varieties. The autumn-sown varieties such as winter oilseed rape, yield more than the spring sown varieties, but the latter grow where winter conditions are too severe for the autumn-

sown varieties. For example, in Canada, where winters are severe, the crop is mainly grown as a spring crop, whereas in Europe, autumn-sown varieties are predominant.

Oilseed rape provides two main products: the oil (from the seeds), and meal (crushed seeds). At present the oil is mainly used for human consumption, though it is used on a small scale as a lubricant and biodiesel fuel. The meal is used as a high-protein feed supplement for livestock and poultry. The advent of varieties with a low eruric acid content in the seventies has improved the quality of the oil for human consumption. The introduction of varieties with low glucosinolate content at the beginning of this decade has improved the quality of the meal for livestock and poultry.

Oilseed rape is the most important oilseed crop of temperate climates. Its oil ranks as third most consumed oil after soybean oil and palm oil and its meal as third most consumed meal after soybean meal and cotton meal. China, EU, India and Canada are the major producers (Verheijen, 1995). Oilseed rape production, like the production of other oilseeds, will have to grow in the coming years to meet increasing demand in the world.

1.2.2 Crop phenological development and seed yield formation

The time of flowering of winter oilseed rape depends on a combined effect of photoperiod and temperature. The low temperature requirement ensures that flower development is delayed until the following spring as photoperiod and temperatures increase in spring. After flowering, temperature is the main determinant of development rate.

Seed yield formation is the result of many interacting processes and their dependence on environmental factors. During the period from sowing in autumn until the onset of flowering in spring, seed yield potential is set by the number of flower buds and leaf area index at onset of flowering (Mendham et al., 1981; Leterme, 1985). Leaf area index peaks around flowering and then declines, and stems and pods take over assimilate production (Brar and Thies, 1977; Chapman et al., 1984; Addo-Quaye et al., 1985). Conditions during flowering are particularly important for pod formation and seed set (Tayo and Morgan, 1975, 1979; Geisler and Henning, 1981; Evans, 1984; Daniels et al., 1986; Leterme, 1985; Keiller and Morgan, 1988). Seed filling is determined either by the sink capacity of the crop, governed by number of seeds and sink capacity of individual seeds, or by the source i.e. assimilate availability during seed filling (Mendham et al., 1981). Remobilization of reserve carbohydrates accumulated during earlier growth phases in roots, stems, leaves and pod hulls may also contribute to seed filling (Rood et al., 1984; Addo-Quay et al., 1985; Quilleré and Tribou, 1987).

2 Model description

2.1 The model

LINTUL-BRASNAP simulates crop growth and development under potential growth conditions with ample supply of water and nutrients in a pest, disease and weed-free environment. The equations for light absorption, crop growth and partitioning of dry matter to the seeds in LINTUL-BRASNAP are presented in Table 2.1 and a schematic representation is given in Fig. 2.1. The symbols used and related acronyms in FORTRAN are described in Appendix II. Parameter values are given in Appendix I.

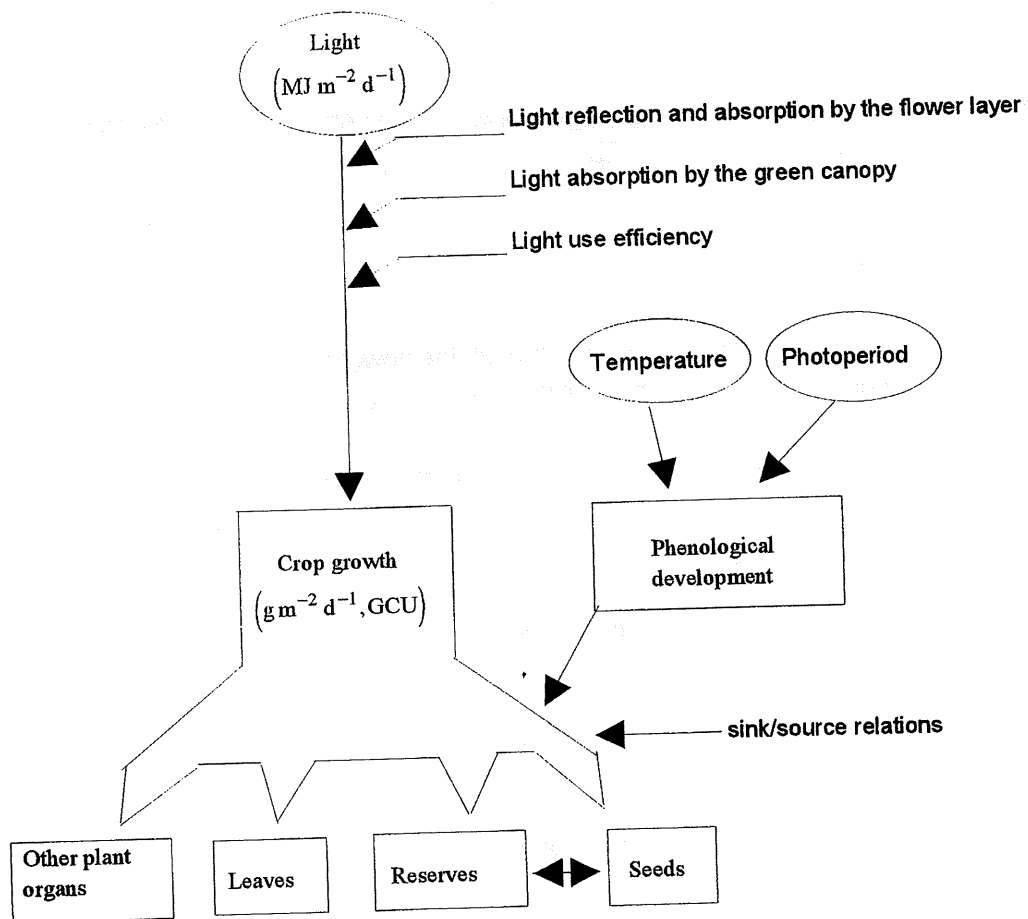


Figure 2.1. Schematic representations of crop growth and partitioning of dry matter of winter oilseed rape as described in LINTUL-BRASNAP.

Table 2.1. Equations of light absorption, crop growth and partitioning of dry matter to the seeds in LINTUL-BRASNAP.

<u>Leaf area</u>		
$LAI = LAI_s \exp(RGR_{LAI} \text{ } tt)$	$0 < LAI \leq 0.75$	1
$LAI = \int (dW/dt) P_{1,DVS} SLA \text{ } dt$	$LAI > 0.75$	2
<u>Flower density, pod density and seed density</u>		
$N_f = f_1 + f_2 W_{tot,OF}$		3
$N_p = p_1 + p_2 W_{tot,EF}$		4
$N_s = s_1 + s_2 \Delta W_{tot,cp}$	$N_s \leq SPP_{pot} N_p$	5
<u>Flower area</u>		
$FAI = N_f A_{if} (D_{if}/D_{fp})$		6
<u>Reflection of light by the flower layer per spectral part a (400-525 nm) or b (525-700 nm) (x)</u>		
$REF_{f,x} = \left(1 - \sqrt{1 - (tr_x + ref_x)}\right) / \left(1 + \sqrt{1 - (tr_x + ref_x)}\right) (1 - \exp(-K_{FAI} \text{ } FAI))$		7
with:		
$K_{FAI} = 0.8 \sqrt{1 - (tr_x + ref_x)}$		8
<u>Transmission of light through the flower area</u>		
$TR = P_a (1 - REF_{f,a}) \exp(-K_{FAI,a} \text{ } FAI) + P_b (1 - REF_{f,b}) \exp(-K_{FAI,b} \text{ } FAI)$		9
<u>Residual light at the bottom of the canopy</u>		
$I_r = I_0 TR (1 - REF_g) \exp(-K_{LAI} \text{ } LAI)$	$1 < DVS \leq 3.5$	10a
$I_r = I_0 (F_{I,r} + RF_{I,r,DVS} (DVS - 3.5))$	$DVS > 3.5$	10b
<u>Light absorption</u>		
$I_a = I_0 TR (1 - REF_g) - I_r$		11
<u>Total crop growth</u>		
$dW_{tot}/dt = LUE_{DVS} I_a$		12
<u>Seed growth</u>		
$dW_s/dt = P_{s,DVS} dW_{tot}/dt - dW_{res}/dt$		13
with: $dW_s/dt \leq R_{is,pot,DVS} (T_{day} - T_{b,4}) N_s$		

Continuation of Table 2.1.

Accumulation or remobilization of reserve-carbohydrates

$$dW_{\text{res}}/dt = -\left[R_{\text{is,pot,DVS}} (T_{\text{day}} - T_{\text{b,4}}) N_s - P_{\text{s,DVS}}(dW_{\text{tot}}/dt)\right] \quad DVS > 3.3 \quad 14$$

with: $dW_{\text{res}}/dt \leq W_{\text{res}} \text{RGR}_{\text{res,max}}$

Crop biomass

$$W_{\text{tot}} = \int_{t_s}^t (dW_{\text{tot}}/dt) dt \quad 15$$

$$W_s = \int_{t_{\text{DVS}=3}}^t (dW_s/dt) dt \quad \text{with: } W_s = W_{\text{is,s}} N_s \quad DVS = 3 \quad 16$$

$$W_{\text{res}} = \int_{t_{\text{DVS}=3.3}}^t (dW_{\text{res}}/dt) dt \quad \text{with: } W_{\text{res}} = F_{\text{res}} W_{\text{tot}} \quad DVS = 3.3 \quad 17$$

Green crop units

$$\text{GRQ}_g = F_r \text{GRQ}_r + F_{\text{st}} \text{GRQ}_{\text{st}} + F_l \text{GRQ}_l \quad 18$$

$$\text{GRQ}_{\text{s,tt}} = R_{\text{foil,tt}} F_{\text{oil,M}} \text{GRQ}_{\text{oil}} + F_{\text{pro,tt}} \text{GRQ}_{\text{pro}} + (1 - R_{\text{oil,tt}} F_{\text{oil,M}} - F_{\text{pro,tt}}) \text{GRQ}_{\text{car}} \quad 19$$

Crop phenological development

$$dDVS/dt = T_{\text{eff}} a_{\text{T,x}} \quad 0 \leq DVS \leq 1 \quad 20a$$

$$= T_{\text{eff}} a_{\text{T,x}} F_v F_p \quad 1 \leq DVS \leq 2; F_v \geq 0; F_p \geq 0 \quad 20b$$

$$= T_{\text{eff}} a_{\text{T,x}} \quad 2 \leq DVS \leq 4 \quad 20c$$

with x: the relevant development phase

$$\begin{aligned} T_{\text{eff}} &= T_{\text{day}} - T_b & T_b < T_{\text{day}} \\ &= 0 & T_b \geq T_{\text{day}} \end{aligned} \quad 21$$

Crop phenological stage

$$DVS(t) = \int_{t_i}^{t_e} dDVS/dt$$

22

with t_i, t_e : beginning and end of a particular development phase, respectively

LINTUL-BRASNAP is an extended version of LINTUL (Spitters, 1990), which is based on Light INTERception and Utilization for total dry matter production, and harvest index for seed yield. The LINTUL model was aimed to extent as far as necessary to explain general trends in seed yield formation under potential growth conditions in West Europe in relation to environmental factors. It has been shown that the harvest index of the crop (seed yield per unit total aboveground dry matter at final harvest) of winter oilseed rape is variable (Mendham et al., 1981; Grosse, 1989). Generally, seed yield is highly correlated with seed density (Mendham et al., 1981; Leterme, 1985; Pouzet et al., 1988; Grosse, 1989) and daily seed growth appears to be determined either by daily assimilate availability (source) or by the maximum growth rate of the seeds (sink) (Mendham et al., 1981; Habekotté, 1993). This is why the partitioning of dry matter to the seeds is not described with a constant harvest index as in LINTUL, but is based on sink- or source-limited growth of the individual seeds (Spitters et al., 1989).

Reflection and absorption of light by the yellow flowers during the period of seed set have been suggested to be important in determining final seed yield (Yates and Steven, 1987; Rao et al., 1991). Therefore, LINTUL was extended to calculate light reflection and absorption by the green leaves and pods and yellow flowers separately, and the total dry matter production is related to light absorption instead of light interception.

LINTUL-BRASNAP also includes accumulation and reallocation of reserve carbohydrates, as they may contribute up to 17.5 % of final seed weight (Rood et al., 1984; Addo-Quay et al., 1985; Quilleré and Triboi, 1987; Habekotté, 1993).

The next paragraphs explain how various crop growth and development processes were modelled in LINTUL-BRASNAP.

2.2 Crop phenological development

Development stage is characterized by *DVS*, a state variable that has the value 0 at sowing, 1 at emergence (E), 2 at the onset of flowering (OF), 3 at the end of flowering (EF) and 4 at maturity (M). The development stage is obtained by integrating of the development rate ($dDVS/dt$) with time steps of one day, according to van Keulen et al. (1982). Development rate is assumed to be linearly related with effective daily temperature (T_{eff}), expressed by the parameter a_T (Thornley, 1987; Spitters et al., 1989). In the period from emergence until the onset of flowering the development rate is also influenced by the degree of vernalization (F_v) and photoperiod (F_p) (Eqs 20-22, Table 2.1).

Effective daily temperature (T_{eff}) for development is calculated from the average day temperature (T_{day}) and a base temperature (T_b) for development (Thornley, 1987) (Eq. 21, Table 2.1). Average day temperature (T_{day}) is calculated as the mean value of daily minimum and maximum temperatures. Vernalization and the effect of photoperiod and vernalization on development rate were described using the approach Weir et al. (1984) and Reinink et al. (1986) applied to winter wheat. The degree of vernalization is represented by a state variable (F_v), with values between 0 (not vernalized) and 1 (fully vernalized) and is calculated by integrating the vernalization rate (dF_v/dt) from emergence until the onset of flowering or until full vernalization. The effect of temperature on the vernalization rate is described in a vernalization temperature response curve (Fig. 2.2a). The multiplication factor for photoperiod (F_p) varies between 0 and 1 and increases linearly between a basal photoperiod (P_b) and a saturating photoperiod (P_{sat}) (Fig. 2.2b) (Habekotté, 1996a).

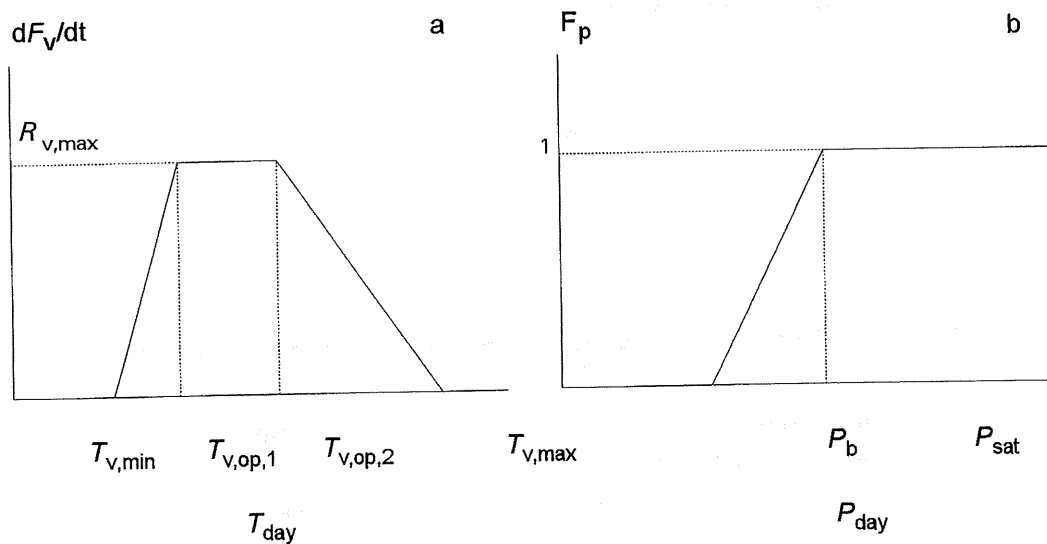


Figure 2.2. Relation between the rate of vernalization (dF_v/dt) and daily temperature (T_{day}) ($T_{v,\min}$: minimum temperature for vernalization, $T_{v,\text{op},1}$: the lowest optimum temperature for vernalization, $T_{v,\text{op},2}$: the highest optimum temperature for vernalization, $T_{v,\max}$: the maximum temperature for vernalization, $R_{v,\max}$: the maximum rate of vernalization) (a). Relation between the multiplication factor for photoperiod effects (F_p) and daily photoperiod (P_{day}) (P_b : basic photoperiod, P_{sat} : saturating photoperiod) (b).

2.3 Light reflection and absorption

Incoming light (I_0 , PAR: photosynthetically active radiation) is partly reflected by the green canopy and, during flowering, part of the I_0 is reflected and absorbed by the yellow flowers. The residual part is absorbed by the green canopy (Eq. 11, Table 2.1) or transmitted to the bottom of the canopy (Eq. 10).

2.3.1 Light reflection and absorption by the flower layer

Transmission (TR) of light by the flower layer is influenced by total flower area (FAl), calculated from the actual number of open flowers and surface area per open flower (A_{if}). Most part of the

flowering period, the flower formation rate proceeds nearly linearly with temperature sum (expressed in degree days) (Fig. 3.2, section 3.1.7). Therefore, the actual density of open flowers was estimated by the ratio of the duration of the flowering per flower and the duration of the flowering period, expressed in degree days (D_{if}/D_{fp}) multiplied by the total number of flowers formed during the flowering period (N_f) (Eq. 6, Table 2.1).

To calculate light transmission through the flower layer, the total spectrum of PAR is separated into two parts i.e. 400-525 nm (a: violet, blue and green light) and 525-700 nm (b: yellow, orange and red light) (Yates and Steven, 1987) and light reflection by the flower layer is calculated separately for both components (Eqs 7 and 8, Table 2.1) according to Goudriaan (1977) (J. Goudriaan, Wageningen Agricultural University (WAU), pers. comm., 1993).

2.3.2 Leaf area development

Leaf area is calculated since the start of regrowth of the crop in spring at t_s . At early stages of crop regrowth, temperatures rather than assimilate availability determine leaf area growth (Eq. 1, Table 2.1). When leaf area index (LAI) exceeds 0.75, the increase in leaf area is calculated as the product of the increase in leaf dry weight and specific leaf area of young leaves (Eq. 2, Table 2.1, Spitters et al., 1989).

2.3.3 Light reflection and absorption by the green canopy

Incoming light or the light transmitted through the flower layer (Eq. 9, Table 2.1) is partly reflected (REF_g) and partly absorbed by the green canopy (Eqs 10 and 11, Spitters et al., 1989). If the LAI is not sufficient for maximum light absorption, the light is also partly transmitted to the bottom of the canopy (Eq. 10). It is assumed that the minimum residual fraction of I_0 at the bottom of the canopy attained at the moment of maximum LAI around flowering remains constant for the remainder of the growing period until just before maturation (Eq. 10a) (Mendham et al., 1981; Habekotté and Smid, 1992). This residual fraction ($RF_{l,r,DVS}$) increases at the end of the growing period with increasing development stage of the crop (Eq. 10b) (Habekotté and Smid, 1992).

2.4 Total dry matter production

LINTUL-BRASNAP relates dry matter production to light absorption by the green parts of the canopy. Daily crop growth rate (dW_{tot}/dt) is calculated from the start of regrowth in spring at t_s as the product of daily light absorption (I_a) and the light use efficiency (LUE_{DVS}) (Eq. 12, Table 2.1). LUE_{DVS} expresses the ratio of dry matter production and light absorption, defined as a function of the developmental stage of the crop (Figure 3.3, Section 3.3). Total crop weight is calculated by integration of daily crop growth rate (Eq. 15). This global approach of crop growth does not include the influence of temperature on crop growth rate.

2.5 Flower density, pod density and seed density

Flower density (N_f) and pod density (N_p) are derived from total dry matter at the onset ($W_{tot,OF}$) and at the end of flowering ($W_{tot,EF}$), respectively (Eqs 3 and 4, Table 2.1; Habekotté, 1993). Seed density (N_s) is derived from total dry matter production during the critical period of seed set (ΔW_{cp}) with a maximum defined as the product of potential number of seeds per pod (SPP_{pot}) and actual pod density (Eq. 5). The critical period of seed set (ΔW_{cp}) is set at 350 °Cd since onset of flowering (Habekotté, 1996b) as explained below.

The duration of the critical period of seed set at crop level was estimated previously as the duration of the flowering period (Leterme, 1985; Habekotté, 1993), i.e. 196 °Cd with a base temperature of 4.9 °C (Habekotté, 1996a). Thus, the duration (in days) of the flowering period strongly depends on temperature regime. Formulated in this way, dry matter production during the flowering period, and thus seed set are also strongly influenced by temperature regime. An alternative hypothesis for the critical period of seed set is defined in this report. This hypothesis is based on three known facts: a) the critical period of seed set per pod is 300 °Cd ($T_b = 0$ °C) since pod initiation (Leterme, 1985), b) from about the end of the critical period of seed set, pod growth becomes increasingly dependent on its own photosynthesis (Leterme, 1985) and c) the influence of temperature is less pronounced on pod initiation than on flower production (own unpublished data, Fig. 2.3). Pod initiation is defined as the moment flowers open (Habekotté, 1993). The duration of the flowering period varies between 250 and 475 °Cd ($T_b = 0$ °C); Leterme, 1985). The difference in temperature response between pod set and flower formation may result in similar durations of the period of pod set (expressed in °Cd) for a range of durations of the flowering period (expressed in °Cd), as illustrated schematically in Figs 2.4a and 2.4b. In earlier research (Habekotté and Smid, 1992) it was found that in flowering periods in the middle of this range, pod initiation was about 50 % at one-third of the flowering period and 75 % at mid-flowering (Fig. 2.4c). The critical period for individual pods (300 °Cd since pod initiation) of 75 % of the initiated pods is indicated by the dashed bold line in Fig. 2.4c. The mean critical period of seed set at crop level was defined based on two conditions: 1) at least 50 % of the critical period of 75 % of all pods initiated is included in the critical period at crop level; 2) the individual critical period of less than 25 % of all pods initiated ends during the critical period at crop level, to minimize interference with independent crop growth. Thus, the critical period at crop level must be between 331 and 361 °Cd. The alternative hypothesis for the duration of the critical period for seed set in a crop is set at 350 °Cd since onset of flowering, for crops with flowering periods of variable length. The duration (d) of the critical period of seed set based on the alternative hypothesis is influenced less by fluctuations in temperature than by the duration of the flowering period, because of a lower base temperature. This hypothesis yielded best simulation results and is integrated in LINTUL-BRASNAP (Habekotté, 1996b).

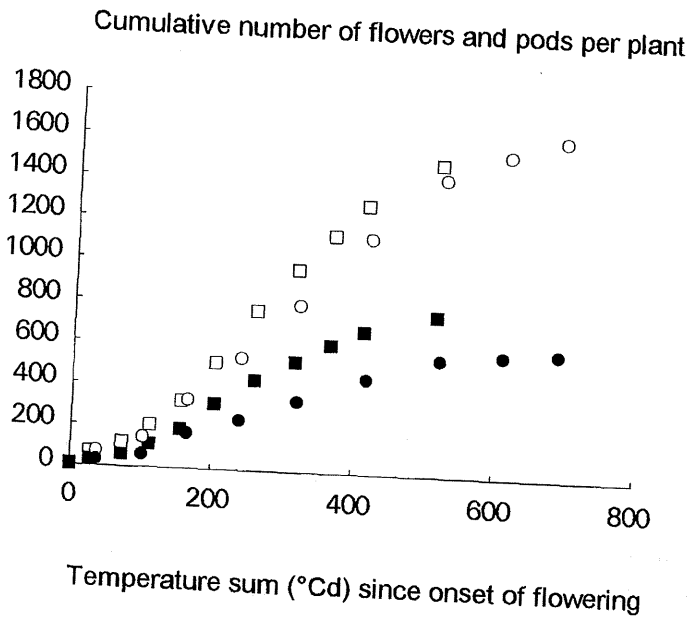
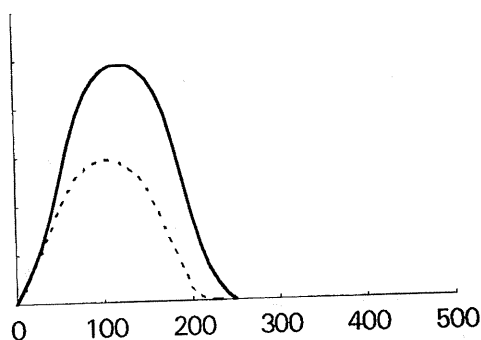


Figure 2.3. Cumulative average number of flowers (open symbols) and pods (solid symbols) per plant (average of six plants) versus temperature sum (°Cd) since onset of flowering based on recorded data of plants of the variety Jet Neuf grown in greenhouses at an average temperature of about 15 °C (squares) and 25 °C (circles).

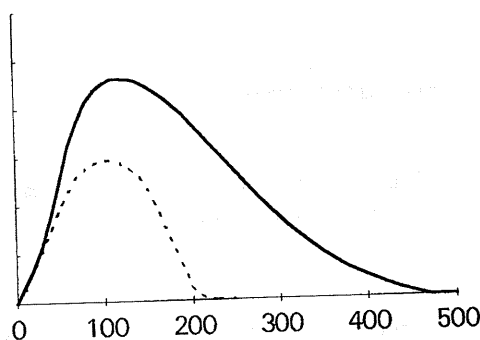
2.6 Seed growth

Considerable accumulation of dry matter in the seeds starts around the end of flowering ($DVS = 3.0$). Since the end of flowering till just after flowering ($DVS = 3.3$) part of the assimilates are distributed to the pod hulls ($1 - P_{s,DVS}$) (Eq. 13, Table 2.1). The partitioning to the seeds gradually increases and after the pod hulls reach their maximum weight (at $DVS = 3.3$) all the assimilates produced by the crop are available for seed growth ($P_{s,DVS} = 1.0$) (Habekotté, 1993). The maximum possible distribution of actually produced assimilates to the seeds (source) is related to the phenological development of the crop and is described by $P_{s,DVS}$. LINTUL-BRASNAP calculates daily seed growth (dW_s/dt) from available carbohydrates for the seeds with a maximum of the potential growth rate per individual seed multiplied by seed density (Eq. 13; Spitters et al., 1989; Habekotté, 1996b). When current assimilates cannot cover the daily growth capacity of the seeds, additional assimilates may be drawn from the reserves up to a maximum depletion rate of the reserve pool ($RGR_{res, max}$) (Habekotté and Smid, 1992). A possible surplus of current assimilates is added to the pool of reserves (Eqs 14 and 19). Total seed weight and the amount of reserves are calculated by integration of daily rates of change (Eqs 16-17).

Flower formation and pod initiation ($(^{\circ}\text{Cd})^{-1}$), a



Flower formation and pod initiation ($(^{\circ}\text{Cd})^{-1}$), b



Cumulative pod initiation (%)

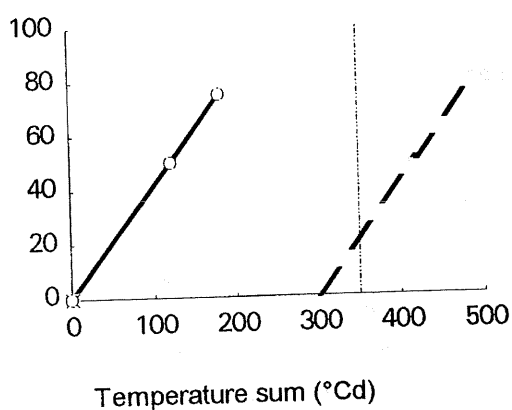


Figure 2.4. Schematic representation of flower formation rate (solid line) and pod initiation rate (dashed line) as a function of temperature sum since the onset of flowering for relatively short (a, 250 $^{\circ}\text{Cd}$, $T_b = 0^{\circ}\text{C}$) and long (b, 475 $^{\circ}\text{Cd}$, $T_b = 0^{\circ}\text{C}$) flowering periods. Average percentages of cumulative pod initiation versus temperature sum (after Habekotté and Smid, 1992) are given for a moderate length of the flowering period (362.5 $^{\circ}\text{Cd}$, $T_b = 0^{\circ}\text{C}$) with a solid bold line (c). The parallel dashed solid line represents the critical period for seed set per initiated pod (solid line plus 300 $^{\circ}\text{Cd}$). The estimated end of the critical period of seed set at crop level is given by the vertical dashed line at 350 $^{\circ}\text{Cd}$ after onset of flowering (c).

2.7 Green crop units (GCU)

Seed weight and reserve carbohydrates are both expressed in weight units similar to those of the green canopy (GCU, g m^{-2} ; green crop units), to allow for differences in chemical composition. The glucose requirement per unit green crop production (GRQ_g in g glucose g^{-1} dry matter) is derived from average fractions of spring and summer production of roots (F_r), stems (F_{st}) and leaves plus pod hulls (F_l) of total dry matter production of the green canopy and from the conversion factors given by Penning de Vries et al., (1989) (Eq. 18), based on the chemical composition of roots, stems and leaves. Glucose requirements for seed production ($\text{GRQ}_{s,tt}$) change with the carbohydrate, protein and oil content of the seeds, defined as a function of thermal time (tt) since pod initiation (Eq. 19). The conversion of seed weight into green crop units is calculated as the ratio $\text{GRQ}_{s,tt}/\text{GRQ}_g$. The conversion of reserve carbohydrates into green crop units is calculated as $1/\text{GRQ}_g$.

2.8 Initial conditions and input data

The initial conditions required for modeling crop development and crop growth are: sowing date in autumn (D_{sow}), initial total crop dry weight ($W_{tot,s}$) and initial leaf area index (LAI_s) at the start of crop growth in spring at t_s .

Input data are daily incoming global radiation (S , $\text{MJ m}^{-2} \text{d}^{-1}$) and daily minimum and maximum temperatures ($T_{\min}$ and $T_{\max}$, respectively). PAR ($\text{MJ m}^{-2} \text{d}^{-1}$) is calculated as $S/2$ and thermal time is calculated by integration of daily effective temperature ($T_{\text{eff}} = ((T_{\min} + T_{\max})/2) - T_b$).

2.9 Model evaluation

The performance of the model was evaluated with data used for parameterization and with other data (Habekotté, 1996a,b). The phenological part of the model was tested with three data sets used for model development and parameterization in the Netherlands, Germany and France, and with two independent data sets in the Netherlands and France, and appeared to be reliable for predicting the onset and end of flowering and the maturity of winter oilseed rape grown on these locations (Habekotté, 1996a). Total model's performance was first evaluated by comparing simulated values of various crop characteristics with recorded data used for parameterization (variety Jet Neuf). The model explained much of the variation in various crop characteristics such as total dry matter production at various stages of development, seed density, pod density and seed yield. When the model's performance was evaluated on an independent data set of early and late sown crops in the UK, observed over a period of 5 years, statistically significant correlations were found between simulated and observed values of various crop characteristics. Differences between sowing dates were simulated better than the differences between years per sowing date. It was concluded that the model recognizably reproduced the observed variation in various crop characteristics (Habekotté, 1996b).

Seed set appeared to be a key factor in the explanation of observed seed yields. Differences in simulated and observed variation in seed yields were attributable to differences in simulated and

observed seed densities. Leterme (1985) and Habekotté (1993; Section 2.5) identified a period identical or nearly identical to the flowering period as the period in which seed density is determined. Habekotté (1993, 1996b) found that seed density could be explained on the basis of total crop dry matter production during this critical period. In the model, the duration of the critical period is determined solely by temperature, and this caused a relatively large variation in simulated seed density compared to observed variation under conditions with maximum light absorption and crop growth rates during flowering (Habekotté, 1996b). The duration (d) of the critical period might be partly governed by temperature and partly by other factors. Keiller and Morgan (1988a) found that in oilseed rape the end of pod and flower production coincides with a sudden cessation of apical growth and sink strength. In a companion paper (Keiller and Morgan, 1988b) they conclude that the pods closer to the base of the plant inhibited the growth and development of the apical regions, possibly because of competition for assimilates or a build-up of inhibitory substances at the apices, or a combination of these. Seed set may be regulated by a mechanism similar to pod set. Further research is needed for a more explanatory and quantitative description of the factors that regulate pod and seed set and that may improve the accuracy of simulating seed density and final seed yield.

The model was evaluated solely with data recorded in the UK and in the Netherlands. The phenology part of the model was also evaluated with data obtained in France and Germany. Further testing of the model is needed to confirm or reject its value for explaining the general trends in seed yield formation in West Europe.

The model simulates the phenological development of the crop from sowing until maturity, and crop growth from early spring until maturity under potential growth conditions. Extending the model to include crop growth during autumn and winter would extend its applicability to locations where initial conditions in spring are not known. Extending it to include crop growth under conditions of limiting water and nutrients would further broaden its applicability.

3 Model parameterization

This chapter describes the quantification of model parameters as far as not described in detail in other publications referred to in Appendix I. Many data are based on the experiments described by Habekotté and Smid (1992). Therefore, these experiments are briefly described in Section 3.1.

3.1 Experiments

The two field experiments used to parameterize LINTUL-BRASNAP were carried out in 1988/89 (experiment 1) and 1989/90 (experiment 2) (Habekotté and Smid, 1992), with variety Jet Neuf on a heavy clay soil near Wageningen, the Netherlands (5° 40' E, 51° 58' N). Details are given in Table 3.1. Treatments were applied to establish variation in total dry matter production and pod density (number of pods per unit area) through varying seed rate, plant density, sowing date, removal of side branches and application of shading. Low plant density was obtained by thinning by hand in autumn and again in early spring. Shading was effected with black plastic netting, from the first week of flowering until final harvest, intercepting 60 % of the light.

Crop development was recorded weekly from early spring until final harvest. Numbers of seeds, pods and seeds per pod, total aboveground dry weight, plant density and the dry weight of roots, green leaves, yellow leaves, main stems, side branches, flower buds, flowers, pod hulls and seeds were recorded at intervals of two or three weeks. Area and weight of green leaves were recorded to determine the specific area (SLA) of the leaves and the area of 5 samples of 50 leaves each were measured to determine mean surface per open flower. Light interception of the crop was determined at one- to two-weekly intervals at noon, by measuring PAR intensity at 5 cm above the soil surface with a portable tube solarimeter.

Reserve carbohydrates (starch and water soluble sugars) were measured in the first experiment, in roots, leaves, stems and pod hulls.

Individual seed growth in marked pods was followed in experiment 1. On three dates during the flowering period of the whole crop (21-4, 28-4 and 3-5), on two plants per sub-plot, coloured wire was attached to the main stem and the first two primary branches above the last open flower. Each week two marked plants were harvested and individual seed weights of the first four pods above each marker recorded.

3.2 Light reflection and absorption

3.2.1 Extinction coefficient for leaves (K_{LAI})

To define the value of the extinction coefficient for leaves (K_{LAI}) (Eq. 10a, Table 2.1) measurements of residual PAR at the bottom of the canopy (I_r) and LAI are necessary for the growth period before onset of flowering. These data were available for the second and third harvest

of the treatments of experiment 1 (Section 3.1). The data of the second treatment were used as these were not influenced by thinning of plant density by hand (treatments 1 and 4) or by cutting side branches (treatment 3) (Table 3.2).

Table 3.1 Details of the field experiments.

Detail	Experiment 1	Experiment 2
Treatments (tr.):	1) Low plant density (lpd) 2) High plant density (hpd) 3) Low plant density and removal at flowering of all side branches except the highest (rsb) 4) Low plant density and shading, leaving 40 % of full light from the first week of flowering till final harvest (sha)	1) Low seed rate (lsr) 2) High seed rate (hsr) 3) Low seed rate and later sowing (lso) 4) Low seed rate and shading (sha, see tr.4, in experiment 1)
Sowing date	: August 23, 1988	August 24, 1989 (early) September 9, 1989 (late)
Sowing rate	: 10 kg ha ⁻¹	6 kg ha ⁻¹ (low seed rate) 10 kg ha ⁻¹ (high seed rate)
Row distance	: 24 cm	24 cm
Plant density in spring	: tr. 1: ca. 24 pl m ⁻² : tr. 2: ca. 68 pl m ⁻² : tr. 3: ca. 23 pl m ⁻² : tr. 4: ca. 27 pl m ⁻²	tr. 1: ca. 42 pl m ⁻² tr. 2: ca. 50 pl m ⁻² tr. 3: ca. 57 pl m ⁻² tr. 4: ca. 41 pl m ⁻²
Date of start shading	: 20-4-1989	24-4-1990
Harvest dates: (day-month)	: 13-3, 28-3, 10-4, 24-4, 8-5, 16-5, 22-5, 29-5, 5-6, 12-6, 19-6, 26-6, 3-7, 10-7	22-3, 18-4, 1-5, 14-5, 28-5, 18-6, 9-7

The K_{LAI} calculated in Table 3.2 is used in LINTUL-BRASNAP (Habekotté, 1996b, c,d). For calculations with a detailed model of canopy photosynthesis based on light distribution within the canopy (DAYASS, Habekotté, 1996d,e) the extinction coefficient for leaves (Table 3.2) was corrected for light interception by stems. This overestimation is estimated to be about 9 %, based on a comparison of a calculation of K_{TAI} based on TAI (the sum of projected areas of stems and leaves; Table 3.3) and of K_{LAI} based on LAI (Table 3.2).

Table 3.2 The extinction coefficient for leaves (K_{LAI}) calculated from residual PAR at the bottom of the canopy in percentages of incoming PAR (I_r / I_0) and LAI (eq. 10, Table 2.1) (Habekotté and Smid, 1992).

Experiment	Treatment	Harvest	Measured	LAI	Calculated
			I_r / I_0 (%)		K_{LAI}
1	2	2	21.6	1.61	0.924
		3	6.6	2.55	1.047
Mean					0.986

Table 3.3 The extinction coefficient for leaves (K_{TAI}) calculated from residual PAR at the bottom of the canopy in percentages of incoming PAR (I_r / I_0) and TAI (eq. 10, Table 2.1) (Habekotté and Smid, 1992). TAI is the sum of the projected areas of stems and leaves.

Experiment	Treatment	Harvest	Measured I_t/I_0 (%)	TAI	Calculated K_{TAI}
1	2	2	21.6	1.70	0.873
		3	6.6	2.86	0.934
Mean					0.903

3.2.2 Relative growth rate of leaves (RGR_{LAI})

The relative growth rate of leaves (RGR_{LAI}) in spring is based on measurements of de Boer and Langenhuis (1985); Table 3.4. In their experiments, values of LAI in spring lower than 0.75 were available (Eq. 1, Table 2.1). The parameterization of RGR_{LAI} is obtained through linear regression of $\ln(LAI / LAI_s)$ versus tt ($RGR_{LAI} = 0.7312$; $R^2 \text{ adj.} = 78.9$)

3.2.3 Specific leaf area (SLA)

The value of the specific leaf area of leaves (SLA) as a basis for calculating the increase in LAI, is based on measurements of young leaves in experiment 1 (Section 3.1) (Table 3.5). The value of SLA increases with harvest date and phenological development stage of the crop. An increase in the values of SLA with development stage of the crop did not improve the simulation results of light absorption and total dry matter production of the treatments in experiment 1 and 2 in comparison with an average value of the SLA. Therefore, the mean value of the SLA was used further in LINTUL-BRASNAP (appendix I).

Table 3.4 Recorded data of the LAI and temperature sum (tt) (de Boer and Langenhuisen, 1985) as a basis for parameterization of RGR_{LAI} (Eq. 1, Table 2.1)

Plant density (m ⁻²)	Sowing	Harvest	Measured <i>LAI</i>	<i>tt</i>	
ca. 80	25-8-1983	22-3-1984	0.54	0	
		11-4-1984	0.70	88.4	
		25-4-1984	1.82	199.5	
	5-9-1983	22-3-1984	0.40	0	
		11-4-1984	0.82	88.4	
		25-4-1984	1.65	199.5	
	15-9-1983	22-3-1984	0.10	0	
		11-4-1984	0.23	88.4	
		25-4-1984	1.09	199.5	
	ca. 40	25-8-1983	22-3-1984	0.37	0
			11-4-1984	0.60	88.4
			25-4-1984	1.24	199.5
5-9-1983		22-3-1984	0.27	0	
		11-4-1984	0.27	88.4	
		25-4-1984	1.19	199.5	

Table 3.5 Measurements of SLA ($cm\ g^{-1}$) of newly formed leaves in four treatments of experiment 1 (Habekotté and Smid, 1992).

Treatment	Harvest date			Mean
	28-3-1989	10-4-1989	24-4-1989	
1	0.0140	0.0177	0.0217	
2	0.0145	0.019	0.0222	
3	0.0141	0.0173	0.0181	
4	0.0136	0.0177	0.0241	
Mean	0.0141	0.0174	0.0215	0.0177

3.2.4 Partitioning of assimilates to the leaves ($P_{I,DVS}$)

The partitioning of assimilates to the leaves is based on experimental data of experiments carried out by Habekotté and Smid (1992) and by de Boer and Langenhuisen (1985). Fig. 3.1 presents the ratio of increase in leaf dry weight and increase in total dry weight (including roots) per development stage of the crop. $DVS0$ represents the interpolated values of the development stage,

based on days. DVS represents the simulated values, based on temperature (including vernalization) and photoperiod (Habekotté, 1996a). Lines were fitted to obtain values of partitioning of assimilates in relation to the development stage of the crop ($DVS0$). The lines were fitted without the values (0,0) in Fig. 3.1. The maximum value of $P_{I,DVS}$ is set at the maximum value of the fitted line (1) at $DVS0 = 1.39$, the stage of development when maximum values of partitioning are obtained in early spring (Habekotté, 1988). The calculated ratio of increase in leaf dry weight and increase in total dry weight (including roots) is the net result of increase in leaf weight through formation of new leaves and of dying off of old leaves. The maximum LAI is mostly reached around flowering ($DVS = 2.0$), when dying off of leaves compensates for the newly formed leaves. This explains the abrupt decrease of $P_{I,DVS}$ at $DVS0 = 2.321$. This is not the result of a gradually decline of formation of new leaves but the result of increase in death of leaves that compensates for the formation of newly formed leaves. The value of the $DVS0$ at this point is based on the value of $DVS0$ in the experiments at the moment that maximum values of the LAI are reached. In LINTUL-BRASNAP, the values of $P_{I,DVS}$ are related to DVS instead of $DVS0$, according the simulated values for experiment 1 (Fig. 3.1).

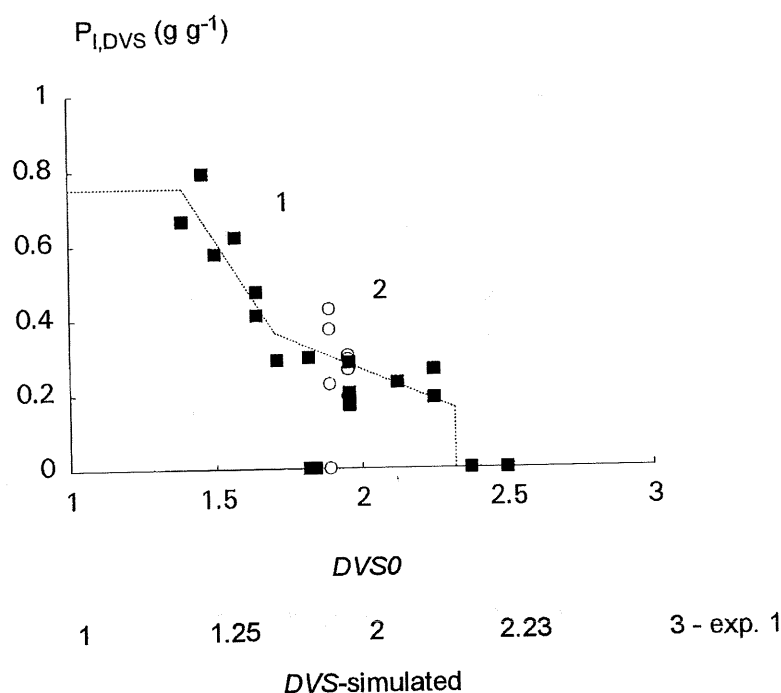


Fig. 3.1 The ratio of increase in leaf dry weight and increase in total dry weight (including roots) versus the interpolated development stage of the crop ($DVS0$). The fitted lines are: $P_{I,DVS} = 2.562 - 1.293 DVS0$, $R^2 \text{ adj.} = 72.0\%$ (1) and $P_{I,DVS} = 0.919 - 0.327 DVS0$, $R^2 \text{ adj.} = 37.1\%$ (2). See text for more details (black symbols: de Boer and Langenhuisen, 1985; open symbols: experiment 1, Habekotté and Smid, 1992).

3.2.5 Fraction of light at the bottom of the canopy near maturity ($RF_{I,r, DVS}$)

It is assumed that the minimum residual fraction of I_0 at the bottom of the canopy attained at the moment of maximum LAI around flowering remains constant for the remainder of the growing period until just before maturation (Eq. 10a) (Mendham et al., 1981; Habekotté and Smid, 1992). This residual fraction ($RF_{I,r,DVS}$) increases at the end of the growing period with increasing

development stage of the crop (Eq. 10b). Parameterization of $RF_{I_r,DVS}$ was based on data of experiment 1 and 2 described by Habekotté and Smid (1992). In experiment 1 the fraction of light at the bottom of the canopy (I_r) was on average 0.073 (as a fraction of incoming light, I_0) at development stage 3.467 and 0.213 at $DVS0 = 4.09$. This resulted in an increase of I_r of 0.225 per unit development stage. In experiment 2 the fraction of light at the bottom of the canopy (I_r) was on average 0.041 (as a fraction of incoming light, I_0) at development stage 3.57 and 0.116 at $DVS0 = 4.1$. This resulted in an increase of I_r of 0.142 per unit development stage. The average value of $RF_{I_r,DVS}$ based on experiment 1 and 2 is 0.184 (Appendix I).

3.2.6 Area of individual flowers (A_{if})

To define the area of individual flowers, the projected area of 5 samples of about 50 flowers were measured. The flowers were taken from plants from the variety Jet Neuf, grown in a greenhouse in 1989 (AB-DLO, data not published). The results are given in Table 3.6 and the mean area of individual flowers was calculated (2.13 cm^2 , Appendix I).

Table 3.6 Projected area of 5 samples of about 50 flowers, and the calculated mean area per individual flower

Sample	Number of flowers	Projected area (cm^2)	Area per flower (cm^2)
1	50	109.4	2.19
2	52	109.8	2.11
3	55	112.8	2.05
4	50	96.1	1.92
5	50	119.5	2.39
Mean			2.13

3.2.7 Flowering duration of individual flowers (D_{if})

The duration of flowering of individual flowers (in degree days) is based on measurements in experiments 1 (Section 3.1). These data were not published in Habekotté and Smid (1992). At five dates during the flowering period of the crop, plants in four treatments and three replicates were marked with coloured wire on the main stem and first side branch, just below the last unopened flower bud and just above the last opened flower. In the following days, the date was recorded when the coloured wire was just under the last opened flower and just above the first formed pod. This date was defined as end of flowering for that particular flower. The average duration of flowering per flower (in days and thermal time) for the five dates mentioned is given in Table 3.7.

Table 3.7 Flowering duration of individual flowers recorded in experiment 1 (Habekotté and Smid, 1992) for five different moments of flower opening during the flowering period of the whole crop (see text). The duration is given in days and in thermal time with a base temperature (T_b) of 0 °C (Habekotté, 1993) and 4.9163 °C (Habekotté, 1996a).

No.	Mean duration of flowering per flower		
	days	thermal time	
		$T_b = 0.0\text{ °C}$	$T_b = 4.9163\text{ °C}$
1	7.5	66.0	29.1
2	6	58.3	28.7
3	4	52.2	32.7
4	5	57.2	32.4
5	12	88.6	29.6
Mean		64.5	30.5

3.2.8 Actual number of open flowers

Both the duration (d) of flowering per flower (Section 3.2.7) and the formation rate of flowers are influenced by temperature. For the largest part of the flowering period, flower formation proceeds nearly linearly with temperature sum; therefore, the actual density of open flowers was estimated by the ratio of the duration (expressed in degree days) of flowering per flower and the duration of the flowering period, (D_{if}/D_{fp}) (Fig. 3.2) multiplied by the total number of flowers formed during the flowering period (N_f) (eq. 6, Appendix I).

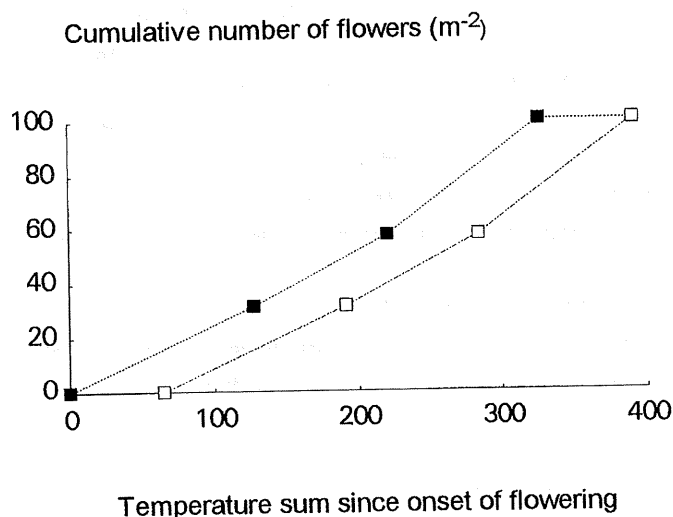


Figure 3.2 Cumulative number of flowers versus temperature sum since onset of flowering in experiment 1 (broken line with solid symbols) (Habekotté and Smid, 1992). The parallel line (broken line with open symbols) represents the duration of flowering per opened flower bud (64.5 °Cd, $T_b = 0.0\text{ °C}$, Section 3.2.7).

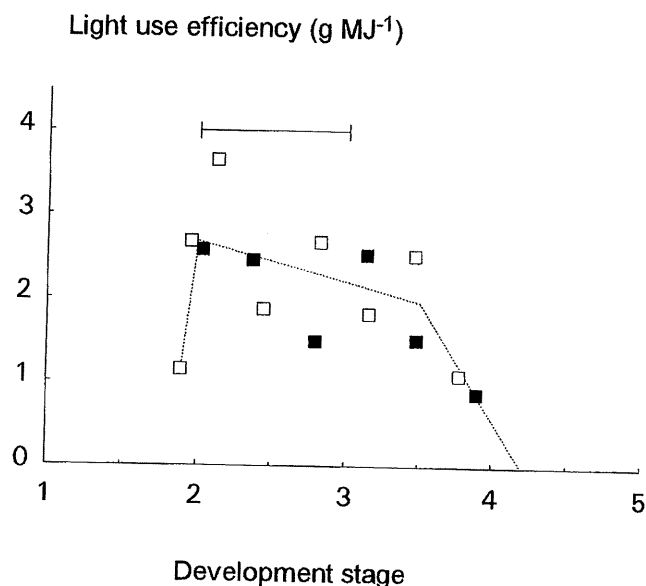


Figure 3.3 Average radiation use efficiency (LUE, g MJ^{-1}) in experiment 1 (open symbols) and experiment 2 (solid symbols) as a function of development stage of the crop (DVS) (Section 3.1). The horizontal bar represents the flowering period.

3.3 Light use efficiency

The light use efficiency, the relation between crop growth (expressed in GCU, Section 3.6) and light absorption was based on the experiments described in Section 3.1. In both experiments the mean values of the light use efficiency for the periods between successive harvests were related to development stage (DVS, Section 2.2) of the crop (Fig. 3.3). Between $DVS = 2$ and 3.5 , LUE showed a slightly decreasing trend with increasing developmental stage of the crop ($P < 0.05$, based on both experiments), but did not differ between the treatments per experiment ($P < 0.05$) or between treatments of both experiments ($t < 0.05$) except for a slightly higher value under shading in experiment 1. The mean value of LUE during this growth period was slightly higher in experiment 1 than in experiment 2 (2.466 g MJ^{-1} compared to 2.173 g MJ^{-1} , respectively, $t < 0.05$). LUE values were low at the beginning of spring and at the end of the crop growth period.

Mendham et al. (1981) also found a lower LUE in early spring, and suggested a relation with low temperatures. In experiments 1 and 2 no relation was found with temperature. The lower value in the beginning of spring may be due to the lower photosynthetic activity of winter grown leaves and a lower LAI at the onset of regrowth (Spitters, 1990; Habekotté and Smid, 1992). At low LAI values the leaves are more exposed to saturating light intensities than at higher LAI_s , resulting in lower light use efficiency in the crop (Habekotté, 1996e).

At the onset of crop growth, the crop's light absorption and dry matter production are governed by leaf area. LAI peaks around flowering and subsequently declines with increasing development stage of the crop and as light absorption and dry matter production are taken over by stems and pods. Some measurements of gross photosynthetic capacities of leaves, stems and pods show that the gross photosynthetic capacities of stems and pods are lower than those of leaves: about 69 % and 38 %, respectively (Rode et al., 1983; de Boer and Langenhuisen, 1985). This may partly explain the gradual reduction in LUE after the onset of flowering in experiments 1 and 2 (Fig. 3.3).

The sharp decline at the end of the growing season ($DVS > 3.5$) may be due to a decline in the photosynthetic capacity of the green canopy with increasing age (Rode et al., 1983; de Boer and Langenhuysen, 1985; Spitters, 1990).

At least 95.3 % (range 95.3-97.2) of the differences in total dry matter production at the onset of flowering, the end of flowering and at maturity of the treatments in both experiments, were explained by LUE_{DVS} . Hence, differences in total dry matter production were largely attributable to cumulative light absorption.

3.4 Seed growth

3.4.1 Partitioning of assimilates to the seeds ($P_{s,DVS}$)

Partitioning of assimilates to the seed already takes place before the end of flowering. However, this amount is low compared to the partitioning after flowering when all the seeds are set. In LINTUL-BRASNAP, partitioning of assimilates to the seeds is based on the experiments described in section 3.1. It is calculated after all seeds are set, i.e. after finishing the critical period of seed set and the flowering period. With the product of individual initial seed weight (W_{is} , Section 3.4.4) and the number of seeds set, an estimation is made of the partitioning of assimilates to the seeds before the end of these periods. Considerable accumulation of dry matter in the seeds start around the end of flowering ($DVS = 3.0$). Since the end of flowering till just after flowering ($DVS = 3.3$) part of the assimilates are distributed to the pod hulls ($1 - P_{s,DVS}$) (Eq. 13, Table 2.1). The partitioning to the seeds gradually increases, and after the pod hulls reach their maximum weight (at $DVS = 3.3$), all the assimilates produced by the crop are available for seed growth ($P_{s,DVS} = 1.0$) (Table 3.8). The maximum possible distribution of actually produced assimilates to the seeds (source) is related to the phenological development of the crop and is described by $P_{s,DVS}$. The data are given in GCU (Sections 2.7 and 3.6)

Table 3.8 Potential partitioning of dry matter (P_s) to the seeds from current dry matter production from just before the end of flowering ($DVS = 2.41$) until just after the end of flowering ($DVS = 3.3$) (Habekotté and Smid, 1992). The data are given in GCU (Section 2.7 and 3.6).

Harvest period	Experiment 1		Experiment 2			Mean	
	DVS	P_s	Harvest period	DVS	P_s	DVS	P_s
H4 - H5	2.45	0	H4 - H5	2.371	0	≤ 2.45	0.0
H5 - H7	2.82	0.0936	H5 - H7	2.79	0.137	2.81	0.115
H7 - H9	3.16	0.849	H7 - H9	3.14	0.216	≥ ca. 3.3	1.0
H9 - H11	3.47	1.04	H9 - H11	3.48	1.44		

3.4.2 Potential growth rate of individual seeds ($R_{is,pot,DVS}$)

LINTUL-BRASNAP calculates daily seed growth (dW_s/dt) from available carbohydrates for the seeds with a maximum of the potential growth rate per individual seed multiplied by seed density

(Eq. 13, Table 2.1; Spitters et al., 1989). The logistic growth curve of individual seeds versus temperature sum since pod initiation (defined as the moment of flower opening; Habekotté, 1993) was used as a starting point to derive the potential growth rates of individual seeds versus the development stage of the crop ($R_{is,pot,DVS}$). The following steps were followed:

- 1) Output of individual seed weight versus temperature sum since pod initiation ($T_b = 0.6870$, basic temperature for development in the seed filling phase, Habekotté, 1996c) based on data of growth of individual seeds with the highest growth rate (treatment 3, period 1 in experiment 1 (Habekotté and Smid, 1992) (Table 3.8).
- 2) Relating the relative oil content to the relative individual seed weight (according to Tribou-Blondel and Ribailles (1988)) and to the temperature sum in step 1 (Table 3.9).
- 3) Calculation of individual seed weight in GCU versus temperature sum since pod initiation, based on oil content (according to step 2), protein content and carbohydrate content versus thermal time since pod initiation (Section 2.7) (Table 3.9).
- 4) Curve fitting (logistic curve) of individual seed weight (in GCU) versus temperature sum ($T_b = 0.687$) since pod initiation based on the data obtained in steps 1-3 ((Table 3.8) ($y = 7.815/(1 + \exp(-0.1196(t - 511.9)))$), $R^2 \text{ adj.} = 98.1$).
- 5) Calculation of the potential growth rate of individual seeds per growth phase of the seeds: a) the exponential growth phase, b) the linear growth phase, and c) the diminishing growth rate. The growth rates in the first and third phase are based on the curve obtained in step 3 (in GCU). The growth rate in the linear growth phase was based on fitting the original data (in GCU) in treatment 3, period 1, experiment 1 (Habekotté and Smid, 1992).
- 6) The potential growth rate versus thermal time was used in LINTUL-BRASNAP since 1/3 of the flowering period (expressed in degree days) when about 50 % of the seed-bearing pods and seeds are set. Output was generated of the potential growth rate of individual seeds, temperature sum and the development stage of the seeds for the growing seasons 1988/89 (experiment 1) and 1989/90 (experiment 2) (Habekotté and Smid, 1992). Based on this output, the potential growth rate was directly related to the development stage of the crop (DVS). Best simulation results were obtained when the potential growth rate was directly related to the development stage of the crop (DVS) in experiment 2, as in this experiment the filling of the seeds was more sink-limited than in experiment 1 (Habekotté, 1993). To obtain optimal simulation results, not all points of the curve in the first and third phase of seed filling were used to parameterize $R_{is,pot,DVS}$. In $R_{is,pot,DVS}$ the potential growth rate is defined per unit thermal time (dW_s/dtt). To obtain the potential growth rate per day (dW_s/dt), $R_{is,pot,DVS}$ is multiplied with the current daily temperature (T_{day}) minus the base temperature for development in the seed filling phase ($T_{b,4} = 0.6780$ °C) (Eq. 13, Table 2.1).

3.4.3 Relative oil content in seeds ($RF_{oil,tt}$)

The relative oil content of individual seeds versus the relative seed weight was derived from Tribou-Blondel and Ribailles (1988) (Table 3.9; variety Jet Neuf). To relate the relative oil content to temperature sum since pod initiation (defined as the moment of flower opening) as a basis to calculate total crop dry weight into GCU (Sections 2.7 and 3.6), a mean logistic growth curve of individual seed weight versus temperature sum was used, based on data of 4 treatments and three periods (Section 3.1) ($y = 4.73/(1 + \exp(-0.00834(t - 539.14)))$), $R^2 \text{ adj.} = 92.1$) (Table 3.9).

Table 3.9 Individual seed weight (in normal dry matter and in GCU, mg) and relative oil content (according to Tribou-Blondel and Ribailier(1988) versus temperature sum since pod initiation ($^{\circ}\text{Cd}$; $T_b = 0.687^{\circ}\text{C}$, basic temperature for development in the seed filling phase, Habekotté, 1996c) based on data of growth of individual seeds with the highest growth rate and (treatment 3, period 1 in experiment 1 (Habekotté and Smid, 1992).

Temperature sum	Individual seed weight		Relative oil content
	normal dry matter	GCU	
0.0	0.0	0.0	0.
440.2	1.826	2.426	0.710
504.9	2.579	3.729	0.927
587.6	3.627	5.387	1
711.8	4.949	7.351	1
828.7	5.454	8.101	1
928.6	4.743	7.045	1
1063.9	5.385	7.998	1

Table 3.10 Temperature sum ($^{\circ}\text{Cd}$) since pod initiation, the relative individual seed weight according a mean logistic growth curve (($y = 4.73/(1 + \exp(-0.00834(tt - 539.14)))$), R^2 adj. = 92.1; see text) and relative oil content versus relative seed weight according to Tribou- Blondel and Ribailier (1988)(all variety Jet Neuf).

Temperature sum since pod initiation	Relative individual seed weight	Relative oil content
0	0	0
275	0.1	0.051
373	0.2	0.41
438	0.3	0.69
490	0.4	0.82
539	0.5	0.94
588	0.6	0.98
641	0.7	1
1200	1	1

3.4.4 Initial individual seed weight ($W_{is,s}$)

The value of initial individual seed weight ($W_{is,s}$) is based on data of total seed weight (in GCU) at the end of flowering ($DVS = 3$) and the final number of seeds in the treatments of experiment 1 and experiment 2 (Section 3.1) (Table 3.11). Seed weight at the end of flowering was obtained by interpolation.

Table 3.11 Initial individual seed weight ($W_{is,s}$) based on data of total seed weight (in GCU) at the end of flowering ($DVS = 3$) and the final number of seeds of experiment 1 and experiment 2 (Habekotté and Smid, 1992).

Experiment	Treatment	Seed weight at the end of flowering (g m ⁻² , GCU)	Final number of seeds (10 ³ m ⁻²)	$W_{is,s}$ (10 ⁻³ g)
Exp. 1	1	30.2	90.4	0.334
	2	40.5	101.6	0.399
	3	14.5	70.8	0.205
	4	9.9	52.1	1.190
Exp. 2	1	17.1	71.0	0.241
	2	22.5	73.8	0.305
	3	15.7	71.8	0.219
	4	5.3	28.4	0.187
Mean				0.254

3.5 Reserve carbohydrates

3.5.1 Fraction reserve carbohydrates (F_{res})

The amount of reserve carbohydrates accumulates until around flowering followed by a decline during the seed filling period (Quilleré and Triboi, 1987; Habekotté, 1993). The maximum amount of reserve carbohydrates is expressed as a fraction of total crop dry weight just after flowering (in GCU) based on the treatments of experiment 1 (Section 3.1) (Table 3.12).

Table 3.12 The amount of reserve carbohydrates as a fraction of total crop dry weight just after flowering in the treatments of experiment 1 (Habekotté and Smid, 1992).

Harvest number	Total crop dry weight (g m ⁻² , GCU)	Sugars and starch (g m ⁻²)	Sugars and starch (g m ⁻² , GCU)	Fraction reserve carbohydrates (GCU)
9	1225	97.5	65.4	0.0534
9	1250	95.1	63.8	0.0511
9	1118	89.5	60.1	0.0537
9	667	38.3	25.7	0.0385
Mean				0.0492

3.5.2 Maximum relative depletion rate of reserve carbohydrates ($RGR_{res,max}$)

LINTUL-BRASNAP calculates daily seed growth (dW_s/dt) from available carbohydrates for the seeds with a maximum of the potential growth rate per individual seed multiplied by seed density (Eq. 13, Table 2.1). When current assimilates cannot cover the daily growth capacity of the seeds, additional assimilates may be drawn from the reserves up to a maximum depletion rate of the reserve pool ($RGR_{res,max}$). Since just after flowering ($DVS = 3.3$), a clear decrease in reserve carbohydrates has been observed (Habekotté, 1993). The use of reserve carbohydrates for seed growth is integrated in the model in the period from just after flowering ($DVS = 3.3$) until maturity. The carbohydrate reserve pool contains starch (at $DVS = 3.3$, less than 10 %) and water soluble sugars. To be mobilized, starch must be hydrolyzed into glucose. This a passive process (it does not require additional energy) (Penning de Vries et al., 1989). Mobilization of glucose requires only a small amount of energy which is neglected in LINTUL-BRASNAP. Redistribution is probable induced when the total demand for sugars exceeds the supply. No information is available on the maximum depletion rate of the reserve pool and whether this rate is related to the amount of actual reserves or not. Without clear experimental evidence, the maximum depletion rate was related to the amount available. Parameterization of $RGR_{res,max}$ is based on data of experiment 1 (Section 3.1; Table 3.13).

Table 3.13 Parameterization of maximum depletion rate of reserve carbohydrates ($RGR_{res,max}$) based on data of experiment 1 described by Habekotté and Smid (1992) ($dW_{res,max}/dt / W_{res} = RGR_{res,max}$).

Treatment	Harvest period	
	H9 - H11	H11 - H14
1	- 0.0524 <---- Maximum value	- 0.0274
2	- 0.0421	- 0.0315
3	- 0.0370	- 0.0377
4	- 0.0309	- 0.0327

3.6 Green crop units

3.6.1 Glucose required for growth of the green canopy (GRQ_g)

Seed weight and reserve carbohydrates are both expressed in weight units similar to those of the green canopy (GCU, $g\ m^{-2}$: green crop units), to allow for differences in chemical composition. The glucose requirement per unit green crop production (GRQ_g in $g\ glucose\ g^{-1}$ dry matter) is derived from average fractions of spring and summer production of roots (F_r), stems (F_{st}) and leaves plus pod hulls (F_l) of total dry matter production of the green canopy and from the conversion factors given by Penning de Vries et al., (1989) (Eq. 18, Table 2.1). The fractions growth of stems (F_{st}),

roots (F_r), leaves and pod hulls (F_l) versus total growth of the green canopy are based on experimental data of experiment 1 (Section 3.1; Table 3.14).

The value of GRQ_g is (Eq. 18, Table 2.1):

$$0.0258 \times 1.444 + 0.557 \times 1.513 + 0.417 \times 1.463 = 1.490 \text{ g g}^{-1}$$

Table 3.14 Growth of stems (F_{st}), roots (F_r), leaves and pod hulls (F_l) as a fraction of total growth of the green canopy in spring and summer, based on experimental data of treatments 1, 2 and 4 in experiment 1 (without treatment 3, because of cutting side branches) described by Habekotté and Smid (1992).

Treatment	ΔW_r (g m ⁻²)	ΔW_{st} (g m ⁻²)	ΔW_l (g m ⁻²)	ΔW_p (g m ⁻²)	$\Delta W_{tot, g}$ (g m ⁻²)	F_r	F_{st}	F_l
1	25.4	472.3	47.4	316.8	861.9	0.0295	0.548	0.423
2	20.1	512.1	67.1	347.4	946.7	0.0212	0.541	0.438
4	14.6	316.3	37.8	175.5	544.2	0.0268	0.581	0.392
Mean						0.0258	0.557	0.417

3.6.2 Glucose required for seed growth (GRQ_s)

The glucose requirements for seed production ($GRQ_{s,tt}$) change with the carbohydrate, protein and oil content of the seeds, defined as a function of thermal time (tt) since pod initiation (Eq. 19, Table 2.1). The relative oil content of individual seeds versus the relative seed weight were derived from Triboi-Blondel (1988) (Table 3.10; variety Jet Neuf). To relate the relative oil content to thermal time as a basis to calculate total crop dry weight into GCU, a mean logistic growth curve of individual seed weight versus thermal time was used, based on data of four treatments and three periods (Section 3.1) ($y = 4.73/(1 + \exp(-0.00834(tt - 539.14)))$, $R^2 \text{ adj.} = 92.1$) (Table 3.10, Section 3.4.3).

For a maximum oil content of 42 % the value of $GRQ_{s,tt}$ is (Eq. 19, Table 2.1):

$$0.42 \times 3.189 + 0.22 \times 1.887 + 0.36 \times 1.275 = 2.214 \text{ g g}^{-1}$$

3.6.3 Green crop units (GCU)

The conversion of seed weight into green crop units is calculated as the ratio $GRQ_{s,tt}/GRQ_g$. For a maximum oil content of 42 % the value of $GRQ_{s,tt}/GRQ_g$ is $2.214/1.490 = 1.485 \text{ g g}^{-1}$. The conversion of reserve carbohydrates into green crop units is calculated as $1/GRQ_g$ ($1/1.490 = 0.671 \text{ g g}^{-1}$).

One weight unit of glucose is equivalent with 0.452 g oilseed weight, 0.671 g GCU and 0.707 g winter wheat grains (Penning de Vries et al., 1989).

4 Use of the model

4.1 FORTRAN Simulation Environment (FSE)

LINTUL-BRASNAP is written in FORTRAN-77 (Meisner and Organick, 1984), using the FSE system (FORTRAN Simulation Environment 2.1, Van Kraalingen (1995)) for crop simulation. The FSE system consists of a main program, a system for weather data input and utilities for specific tasks. The subroutine SUBNAP contains the actual model equations described in Chapter 2.

One of the main features of FSE is the distinction of four tasks that control the order of calculations in the subroutine SUBNAP: ITASK = 1 for initialization; ITASK = 2 for rate calculations; ITASK = 3 for state calculations/updates; and ITASK = 4 to mark the end of the program. The FSE environment retrieves weather data from the WEATHER system as described by Van Kraalingen et al. (1991), enables input of parameters and initial states and has the capability to carry out reruns on these parameter values. Facilities are provided for the output of simulation results in the form of tables or graphs, and time control. The FSE system uses utility routines from the TTUTIL utility library (Rappoldt and Van Kraalingen, 1990). The FSE program can easily be run in different computer environments. The report "The FSE system for crop simulation 2.1" (Van Kraalingen, 1995) describes in detail the FSE system. It supplies guidelines for a crop modeller to make his routines compatible with FSE and it contains a Users Guide, which describes how to operate a FSE model and its data files.

The FSE source program LINTUL-BRASNAP is distributed on floppy disk with the necessary libraries to compile, link and run the program if you are working with Microsoft Fortran 5.1. The full TTUTIL library and WEATHER system can also be obtained on separate floppy disks. Write or e-mail to the address below.

Research Institute for Agrobiolgy and Soil Fertility (AB-DLO)
Software Product Support
P.O. Box 14,
6700 AA Wageningen,
The Netherlands.
e-mail: sps@ab.agro.nl

4.2 Components of LINTUL_BRASNAP

Fig. 4.1 presents a layout of the components of LINTUL-BRASNAP. The execution starts with the MAIN program. This is, however, a one-line program with nothing more than a single call to the FSE-driver. The FSE-driver takes care of task sequencing (e.g. initialization, rate and state calculation, output timing), retrieves and checks weather data from file, takes care of correct integration of various scientific submodels that may be present (in LINTUL-BRASNAP only the subroutine SUBNAP, which is the actual growth routine), reading input and output file names from the input file CONTROL.DAT, the use of reruns, reading time control, output control and weather control variable from the timer file and creation of output tables in the right format. To do so, the FSE driver makes calls to the WEATHER system for weather data control, and to the utility library

TTUTIL. In addition, calls to TTUTIL may also be made in the scientific model subroutine (i.e. SUBNAP) for tasks such as input data reading and output data writing.

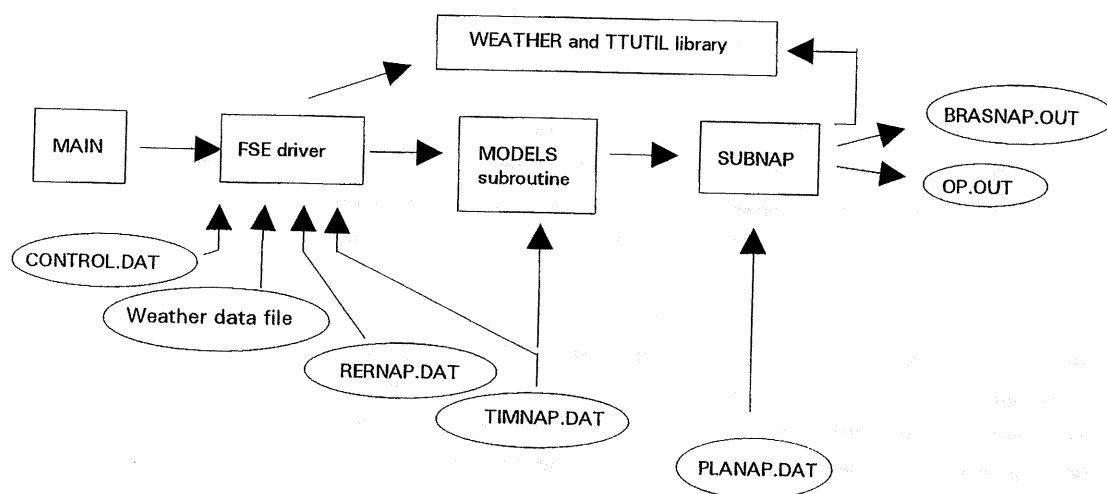


Figure 4.1 Layout of the components of LINTUL-BRASNAP in the FSE system (After van Kraalingen, 1995). SUBNAP contains the actual crop growth routine explained in this report.

Table 4.1 presents an overview of source code files, libraries and standard input data files for LINTUL-BRASNAP. The names of these input data files are specified in the file CONTROL.DAT. Complete listings of these files are given in Appendix III.

4.3 Use of the model with the standard parameter set

4.3.1 Initial conditions and input data

The model does not require interactive input during execution. The runs have been specified completely in the data files. The initial conditions required for modelling crop development and crop growth are: year of sowing (IYEAR), sowing date in autumn (SWD, D_{sow}), initial total crop dry weight (TDW, $W_{tot,s}$) and initial leaf area index (SLAI, LAI_s) at the start of crop growth in spring at t_s (STIM). These data have to be supplied by the user (see Table 4.2).

LINTUL-BRASNAP requires data on daily incoming global radiation (S , $\text{MJ m}^{-2} \text{d}^{-1}$) and daily minimum and maximum temperatures (T_{min} and T_{max} , respectively).

Table 4.2 presents an overview in which input files the initial conditions have to be filled in by the user. Incoming global radiation and daily minimum and maximum temperatures are obtained from the weather data files. Appendix III gives an example weather file.

Other (crop) parameters as given in PLANAP.DAT should only be changed by the user based on new insights or based on additional experimental evidence.

Table 4.1 Source code files, libraries and standard input data files for LINTUL-BRASNAP.

Standard name in FSE	Name for LINTUL-BRASNAP	Explanation/contents
<u>Source code</u>		
Models	Models	Interface routine between FSE driver and the actual simulation model
FSE	FSE	FSE driver
MODEL	SUBNAP	model of crop growth and phenological development of winter oilseed rape
RDDATA	RDDATA	Utility routines for data reading
<u>Object libraries</u>		
WEATHER	WEATHER	WEATHER system
TTUTIL	TTUTIL	Utility subroutines, e.g. for data input and output
<u>Input files</u>		
CONTROL.DAT	CONTROL.DAT	Names of input and output files to be used (e.g. TIMNAP, PLANAP etc....)
TIMER.DAT	TIMNAP.DAT	Time variables (year, start time, etc.), weather station, country, output control variables.
PLANT.DAT	PLANAP.DAT	Plant parameters and initial values of state variables
RERUNS.DAT (Optional)	RERNAP.DAT	Defines reruns: parameter from input files (timer, plant), or name(s) of plant and timer data files
Weather data files		Daily weather data
<u>Output files</u>		
RESULTS.OUT	BRASNAP.OUT	File with output data per time period
	OP.OUT	PRDEL Extra output file with data at maturity

Table 4.2 Input data required for simulation runs with LINTUL-BRASNAP.

Input data/ acroym	Explanation	Input files for LINTUL-BRASNAP
IYEAR	Year of sowing	TIMNAP.DAT
CNTR	Country of weather data	"
ISTN	Station number of weather data	"
STTIME	Start day of simulation (must be earlier than SWD)	"
FINTIM	Finish time of simulation (must be later than maturity)	"
SWD	Sowing data	PLANAP.DAT
STIM	Start day for integration of crop growth in spring	"
TDW	Initial value of total crop weight in spring (including roots)	"
SLAI	Initial value of LAI in spring	"
<u>Optional:</u>		
TDW_OBS	Observed values of TDW, SEEDW and/or LAI during the	"
SEEDW_OBS	growing season (the harvest dates have to be filled in also in	"
LAI_OBS	TIMNAP.dat).	"
All changed initial values for the reruns have to be placed in this input file rernap.dat		RERNAP.DAT

4.3.2 Output of data

Table 4.3 shows how the output of the model can be managed. It is also possible to view the output graphically on IBM-PC's and compatibles. This can be done with the TTSELECT program, provided DELTMP is set to 'N' in the timer file. TTSELECT, however, is not part of the FSE standard distribution software. For more detail the reader is referred to van Kraalingen (1995).

It is possible to include experimental data of various variables such as total dry matter production, weight of leaves, stems and seeds and LAI in the input file PLANAP.DAT for comparison with simulation results in the output file and figures.

The experimental data have to be filled in in the arrays TDW_OBS, LAI_OBS e.g. Observations are given in the sequence <year of observation>, <day number of observation>, <observed value>. As an example for LAI:

```
LAI_OBS =
1989., 72., 0.79,
1989., 87., 1.04,
1989., 100., 1.79,
1989., 114., 2.42,
1989., 128., 2.03,
1989., 142., 1.02,
1989., 156., 1.00,
1989., 170., 0.12,
```

1989., 184., 0.02,
1989., 191., 0.03

When no observed data are available, the value -99 should be given.

Table 4.3 Management of output of data of LINTUL-BRASNAP.

Acronym	Acronym in:	Explanation	Output to file
OPSTOR	SUBNAP	Fill in the wanted output data at maturity, such as TDW, SEEDW, DOF, DEF etc.	OP.OUT
PRDEL	TIMNAP.DAT	Standard time between consecutive outputs to file: BRASNAP.DAT and for graphical output.	BRASNAP.OUT
<u>Optional:</u>			
IOBSD		Fill in list of output-day data for which output is requested; for example at special harvest data corresponding with TDW_OBS etc.	BRASNAP.OUT
STRUN	CONTROL.DAT	Give the run numbers for which you want output, strun=0 is the default run	OP.OUT
ENDRUN			BRASNAP.OUT

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Appendix I: Parameter values

Table I.1: Parameter values in LINTUL-BRASNA.

Parameter	Value	Unit	Source
Phenological development			
$T_{b,1}$	0.3024	°C	Habekotté (1996a)
$T_{b,2}$	0.5444	°C	"
$T_{b,3}$	4.9163	°C	"
$T_{b,4}$	0.6870	°C	"
$a_{t,1}$	7.7212×10^{-3}	d ⁻¹ °C ⁻¹	"
$a_{t,2}$	2.0083×10^{-3}	d ⁻¹ °C ⁻¹	"
$a_{t,3}$	5.1036×10^{-3}	d ⁻¹ °C ⁻¹	"
$a_{t,4}$	1.4651×10^{-3}	d ⁻¹ °C ⁻¹	"
$T_{v,min}$	-3.7182	°C	"
$T_{v,op,1}$	0.7260	°C	"
$T_{v,op,2}$	5.3770	°C	"
$T_{v,max}$	17.2022	°C	"
$R_{v,max}$	0.014553	d ⁻¹ °C ⁻¹	"
P_b	5.7416	h	"
P_{sat}	14.8014	h	"
Crop growth and light absorption			
LUE_{DVS}	0.000, 0.000, 0.999, 0.000, (at t_s), 1.145, 2.000, 2.682, 3.500, 1.957, 4.200, 0.000	g MJ ⁻¹	Habekotté and Smid, 1992; this report
K_{LAI}	0.986		Habekotté and Smid, 1992; this report
RGR_{LAI}	0.00732	°Cd ⁻¹	de Boer and Langenhuisen, 1985; this report
SLA	0.0177	cm ² g ⁻¹	Habekotté and Smid, 1992; this report
$P_{l,DVS}$	0.000, 0.000, 1.000, 0.752, 1.125, 0.752, 1.425, 0.363, 1.580, 0.330, 1.780, 0.298, 2.010, 0.265, 2.321, 0.160, 2.32101, 0.0, 6.000, 0.000		Habekotté and Smid, 1992; de Boer and Langenhuisen, 1985; this report
$RF_{l,r,DVS}$	0.184		Habekotté and Smid, 1992; this report
REF_g	0.046 before OF 0.072 after EF		Yates and Steven, 1987 and Leach, 1987;
A_{if}	2.13	cm ²	Habekotté and Smid, 1992 (this report)

Continuation of Table I.1.

Parameter	Value	Unit	Source
D_{if}	30.5 ($T_b=4.9163$ °C)	°Cd	Habekotté and Smid, 1992 (this report)
D_{fp}	195.94 ($T_b=4.9163$ °C)	°Cd	Habekotté, 1996a
ref_a ; ref_b	0.035; 0.450		Steven and Yates, 1987
tr_a ; tr_b	0.035; 0.150		"
P_a ; P_b	0.400; 0.600		"
flower, pod and seed density			
f_1 ; f_2	6380; 7.34.	m^{-2}	Habekotté, 1993
p_1 ; p_2	-1021.; 6.76	m^{-2}	"
s_1 ; s_2	-7663.; 184.1	m^{-2}	Habekotté, 1996b
SPP_{pot}	35		Mendham et al., 1984
seed growth			
$P_{s,DVS}$	0.00, 0.000, 2.41, 0.000, 2.81, 0.115, 3.30, 1.000, 5.00, 1.000	$g\ g^{-1}$	Habekotté and Smid, 1992; this report
$W_{is,s}$	0.254×10^{-3}	g	"
$R_{is,pot,DVS}$	0.000, 0.000, 2.999, 0.000, 3.000, 2.838, 3.110, 6.346, 3.256, 15.386, 3.330, 20.086, 3.505, 20.086, 3.549, 17.917, 3.769, 4.841, 4.355, 0.045, 6.000, 0.000	10^{-3} $g\ ^\circ Cd^{-1}$	"
reserve carbohydrates			
F_{res}	0.049	$g\ g^{-1}$	Habekotté and Smid, 1992; this report
$RGR_{res,max}$	-0.0524	$g\ g^{-1}\ d^{-1}$	Habekotté and Smid, 1992; this report
Green crop units (GCU)			
F_r ; F_{st} ; F_l	0.026; 0.557; 0.417	$g\ g^{-1}$	Habekotté and Smid, 1992; this report
$RF_{oil,tt}$	0.0, 0.00, 275., 0.051, 373., 0.41, 438., 0.69, 490., 0.82, 539., 0.94, 588., 0.98, 641., 1.00, 1200., 1.00	$g\ g^{-1}$	Triboi-Blondel and Ribailier, 1988; this report
$F_{oil,M}$	0.42 (1988/89, 1989/90) 0.43 (1981-90, all sowings) 0.43 (1975-89, first sowings)	$g\ g^{-1}$	van der Meulen, 1990a,b
$F_{pro,tt}$	0.00, 0.22, 1200., 0.22	$g\ g^{-1}$	Triboi-Blondel and Ribailier, 1988
$F_{car,tt}$	$= 1 - F_{oil,tt} - F_{pro,tt}$	$g\ g^{-1}$	calculated
$GRQ_{r,...,st,...,l}$	1.444; 1.513; 1.463	$g\ g^{-1}$	Penning de Vries et al., 1989
$GRQ_{o,...,p,...,c}$	3.189; 1.887; 1.275		
$T_{b, is}$	0.0	°C	Habekotté, 1993

Appendix II: Acronyms and symbols

II-a: Acronyms and symbols

Table II.1: Acronyms in LINTUL-BRASNAP (local in SUBNAP) (S: sowing, E: emergence, OF: onset of flowering, M: maturity).

Acronym	Variable type	Symbol	Dimension	Description
ALFA	auxiliary		g MJ^{-1}	
AT1	parameter	$a_{T,1}$	$\text{d}^{-1} \text{ }^{\circ}\text{C}^{-1}$	Response of the development rate to temperature for the period from S till E
AT2	parameter	$a_{T,2}$	$\text{d}^{-1} \text{ }^{\circ}\text{C}^{-1}$	Response of the development rate to temperature for the period from E till OF
AT3	parameter	$a_{T,3}$	$\text{d}^{-1} \text{ }^{\circ}\text{C}^{-1}$	Response of the development rate to temperature for the period from OF till EF
AT4	parameter	$a_{T,4}$	$\text{d}^{-1} \text{ }^{\circ}\text{C}^{-1}$	Response of the development rate to temperature for the period from EF till M
CONVG	parameter	$(\text{GRQ}_g)^{-1}$	g g^{-1}	Conversion factor of glucose units into green crop units dry matter
D350	auxiliary		d	Julian day number at 350 °Cd after OF
DAYLP	auxiliary	P	h	Photoperiodic daylength
DEF	output		d	Julian day number at the end of flowering
DEMAND	auxiliary		g m^{-2}	Demand for reserve carbohydrates by the seeds (GCU)
DFF	output		d	Julian day number at full flowering (OF + 7 d)
DGR	rate		d d^{-1}	Rate variable with value 1. for calculation of DGRAF
DGRAF	auxiliary		d	Number of days after 1-1 (for making figures)
DLAI	rate	$d\text{LAI}/dt$	$\text{m}^2 \text{ m}^{-2} \text{ d}^{-1}$	Daily increment of the LAI
DLB	parameter	P_b	h	Basal photoperiod for development
DLE	auxiliary	P_{eff}	h	Effective photoperiod for development
DLF	auxiliary	F_p		Multiplication factor for the influence of photoperiod on the development rate
DLOPT	parameter	P_{sat}	h	Saturating photoperiod
DOF	output		d	Julian day number at OF
DOYM	output		d	Julian day number at M
DPFL	parameter	D_{if}	$^{\circ}\text{Cd}$	Duration of flowering per flower
DRECAR	rate variable	dW_{res}/dt	g m^{-2}	Daily increment of reserve carbohydrates (GCU)
DSEEDW	rate variable	dW_s/dt	g m^{-2}	Daily increment of seed weight (GCU)
DTDW	rate variable	dW_{tot}/dt	g m^{-2}	Daily increment of total crop dry weight (GCU)

Continuation of Table II.1.

Acronym	Variable type	Symbol	Dimension	Description
DVR	rate variable	$dDVS/dt$	d^{-1}	Daily development rate
DVS	state variable	DVS		State of development
DVSLUE	auxiliary			State of development at the onset of crop growth early in spring, used for calculation of the RUE before flowering
DVSM	output			State of development at maturity; for output
FAI	auxiliary	FAI	$m^2 m^{-2}$	Flower area index
FIX	parameter		d	Start day of simulation; default value 220.
FL	auxiliary	P_l	$g g^{-1}$	Partitioning of daily crop growth to the leaves
FLDVS	function	$P_{l,DVS}$	$g g^{-1}$	Partitioning of daily crop growth to the leaves, related to the development stage of the crop
FOILM	parameter	$F_{oil,M}$	$g g^{-1}$	Oil content of the seeds at maturity
FPAR	auxiliary	$F_{l,r}$	$MJ MJ^{-1}$	Fraction of incoming PAR at the bottom of the canopy
FPAR35	auxiliary	$F_{l,r,3.5}$	$MJ MJ^{-1}$	Fraction of incoming PAR at the bottom of the canopy when the development stage is 3.5 (near maturity)
FPDVS	parameter	$RF_{l,r,DVS}$		Response of the fraction of PAR at the bottom of the canopy near maturity ($DVS > 3.5$) to the increment of the development stage of the crop
FPER	parameter	D_{fp}	$^{\circ}Cd$	Average duration of the flowering period
FPROT	parameter	F_{pro}	$g g^{-1}$	Protein content of the seeds during the seed filling phase
FRECAR	parameter	F_{res}	$g g^{-1}$	Fraction of reserve carbohydrates of total crop dry matter, just after flowering at $DVS=3.3$ (GCU)
FS	auxiliary	P_s	$g g^{-1}$	Partitioning of daily crop growth to the seeds
FSDVS	function	$P_{s,DVS}$	$g g^{-1}$	Partitioning of daily crop growth to the seeds, related to the development stage of the crop
GC	parameter	GRQ_{car}	$g g^{-1}$	Glucose units required for formation of carbohydrates
GO	parameter	GRQ_{oil}	$g g^{-1}$	Glucose units required for formation of oil
GP	parameter	GRQ_{pro}	$g g^{-1}$	Glucose units required for formation of protein
GRA	auxiliary	$REF_{f,a}$	$MJ MJ^{-1}$	Reflection of incoming spectral part a of PAR (400-525 nm), by the flower layer
GRB	auxiliary	$REF_{f,b}$	$MJ MJ^{-1}$	Reflection of incoming spectral part b of PAR (525-700 nm), by the flower layer
GRQS	auxiliary	GRQ_s	$g g^{-1}$	Glucose units required for formation of seed dry matter
HI1	output		$g g^{-1}$	Harvest index based on seed weight/total top crop dry weight (GCU)
HI2	output		$g g^{-1}$	Harvest index based on seed weight/total top crop dry weight (normal dry matter)

Continuation of Table II.1.

Acronym	Variable type	Symbol	Dimension	Description
IMSW	parameter	$W_{is,s}$	10^{-3} g	Initial individual seed weight (GCU per 1000 seeds) at the end of flowering (at DVS=3.0)
IT3,4,7,8,11,12	parameters			Parameter for definition of arrays
KLAI	parameter	K_{LAI}		Extinction coefficient for leaves
L2,3 and 6	parameters			Parameter for definition of arrays
LAI	state variable	LAI	$m\ m^{-2}$	Leave area index
LAIM	output		$m\ m^{-2}$	Maximum leave area index
LUE	auxiliary	LUE	$g\ MJ^{-1}$	Light use efficiency
LUEDVS	parameter (i)	LUE_{DVS}	$g\ MJ^{-1}$	Light use efficiency related to the development stage of the crop
LUEM	auxiliary		$g\ MJ^{-1}$	Radiation use efficiency
MDREC	parameter	$RGR_{res,max}$	$g\ g^{-1}\ d^{-1}$	Maximum depletion rate of the pool of reserve carbohydrates (GCU)
MSWM	output		g	Mean seed weight at maturity (per 1000 seeds GCU); only for output
MVR	parameter	$R_{v,max}$	d^{-1}	Maximum vernalization rate
NOFL	auxiliary	N_f	m^{-2}	Flower density
NP1	parameter	P_1	m^{-2}	Parameter for calculation pod density
NP2	parameter	P_2	m^{-2}	Parameter for calculation pod density
NPOD	auxiliary	N_p	m^{-2}	Pod density
NS1	parameter	S_1	m^{-2}	Parameter for calculation seed density
NS2	parameter	S_2	m^{-2}	Parameter for calculation seed density
NSEED	auxiliary	N_s	m^{-2}	Seed density
NSNP	auxiliary			Number of seeds per pod
OPT1,2,3,4	auxiliary		$^{\circ}C$	Maximum daily temperature for crop development (dummy values)
PAR	forcing	I_0	$MJ\ d^{-1}$	Daily incoming PAR (photosynthetic active radiation)
PARA	auxiliary	I_a	$MJ\ d^{-1}$	Daily absorbed PAR (photosynthetic active radiation) by the green crop canopy
PARR	forcing	I_r	$MJ\ d^{-1}$	Daily PAR at the bottom of the canopy (photosynthetic active radiation)
PDMSW	auxiliary	$R_{is,pot}$	$g\ ^{\circ}Cd$	Daily potential individual seed growth per 1000. seeds (GCU)
PDMSWT	function	$R_{is,pot,DVS}$	$g\ ^{\circ}Cd$	Daily potential individual seed growth per 1000. seeds (GCU), related to the development stage of the crop
PF1	parameter	F_1	m^{-2}	Parameter for calculation flower density
PF2	parameter	F_2	m^{-2}	Parameter for calculation flower density
PSEEDW	auxiliary		$g\ m^{-2}\ d^{-1}$	Potential daily seed growth (GCU)
PSPP	parameter	SPP_{pot}		Potential number of seeds per pod
RCALL,1-9	auxiliary			Auxiliary variables for calculations
RECAR	state variable	W_{res}	$g\ m^{-2}$	Weight of reserve carbohydrates (GCU)

Continuation of Table II.1.

Acronym	Variable type	Symbol	Dimension	Description
REFGAF	parameter	REF_g	$MJ\ MJ^{-1}$	Reflection of PAR by the green canopy after the flowering period
REFGBF	parameter	REF_g	$MJ\ MJ^{-1}$	Reflection of PAR by the green canopy before the flowering period
REFLC	auxiliary	REF_g	$MJ\ MJ^{-1}$	Reflection of PAR by the green canopy
RLAI	parameter	RGR_{LAI}	$m^2\ m^{-2}\ ^\circ Cd^{-1}$	Relative growth rate of the LAI with thermal time (if $LAI < 0.75$)
ROILTB	function	$F_{oil,tt}$	$g\ g^{-1}$	Oil content of the seeds related to the temperature sum since 1/3 of the flowering period (since $DVS=2.333$)
SEEDW	state variable	W_s	$g\ m^{-2}$	Seed weight (GCU)
SEWM	output		$g\ m^{-2}$	Seed weight (GCU) at maturity; only for output
SLA	parameter	SLA	$m^2\ g^{-1}$	Specific leaf area of young leaves
SLAI	initial state	LAI_s	$m^2\ m^{-2}$	Initial leaf area index at the start of regrowth of the crop in early spring
SMSW	auxiliary	$W_{s,i}$	$g\ m^{-2}$	Initial seed weight at the end of flowering
SPFL	parameter	A_{if}	m^2	Aea per individual flower
STIM	parameter	t_s	d	Time of start of regrowth of the crop in spring (julian day number plus 356. or plus 366. (leap-year)
SWD	parameter	D_{sow}	d	Sowing day (Julian day)
SWNM	output		$g\ m^{-2}$	Seed weight at maturity (normal dry matter); only for output
SWNOR	auxiliary		$g\ m^{-2}$	Seed weight (normal dry matter)
TB1	parameter	$T_{b,1}$	$^\circ C$	Basic temperature for development in the period from S till E
TB2	parameter	$T_{b,2}$	$^\circ C$	Basic temperature for development in the period from E till OF
TB3	parameter	$T_{b,3}$	$^\circ C$	Basic temperature for development in the period from OF till EF
TB4	parameter	$T_{b,4}$	$^\circ C$	Basic temperature for development in the period from EF till M
TDW	state variable	W_{tot}	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU)
TDW350	auxiliary	W_{350}	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at 350 $^\circ Cd$ after OF
TDWEF	auxiliary	W_{EF}	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at EF
TDWFF	output	W_{FF}	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at full flowering (=OF plus 7 d)
TDWM	output	W_M	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at M
TDWNOR	auxiliary		$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; normal dry matter)

Continuation of Table II.1.

Acronym	Variable type	Symbol	Dimension	Description
TWNT	auxiliary		g m^{-2}	Total top crop dry weight (including dead leaves and roots; normal dry matter)
TDWOF	auxiliary	W_{OF}	g m^{-2}	Total crop dry weight (including dead leaves and roots; GCU) OF
TDWTOP	output	W_{top}	g m^{-2}	Total top crop dry weight (including dead leaves and roots; GCU)
TINY	contant			Constant with value: 10^{-4}
TLAI	auxiliary			Base temperature for calculation of the temperature sum since the onset of crop regrowth in spring for calculation of LAI development ($\text{LAI} < 0.75$)
TMAX	parameter	$T_{v,\text{max}}$	$^{\circ}\text{C}$	Maximum effective temperature for vernalization
TME	auxiliary	T_{eff}	$^{\circ}\text{C}$	The effective day temperature for crop development
TMG	auxiliary	T_{day}	$^{\circ}\text{C}$	Average daily temperature
TMIN	parameter	$T_{v,\text{min}}$	$^{\circ}\text{C}$	Minimum effective temperature for vernalization
TMM	auxiliary		$^{\circ}\text{C}$	Daily effective temperature for calculation a temperature sum since the start of regrowth in spring, used for calculation of LAI development
TMS	parameter		$^{\circ}\text{C}$	Effective temperature for calculation daily potential seed growth
TOF	auxiliary			Base temperature for calculation of the temperature sum since the onset of flowering for calculation of the critical period of seed set
TOP1	parameter	$T_{v,\text{op},1}$	$^{\circ}\text{C}$	Lowest optimum temperature for vernalization
TOP2	parameter	$T_{v,\text{op},2}$	$^{\circ}\text{C}$	Highest optimum temperature for vernalization
TPARA	state		MJ	Cumulative absorbed PAR since the start of regrowth in spring
TRANFP	output	TR	MJ MJ^{-1}	Transmission of PAR through the flower layer; only for output
TRANS	auxiliary	TR	MJ MJ^{-1}	Transmission of PAR through the flower layer
TRANSA	auxiliary	TR_a	MJ MJ^{-1}	Transmission of PAR through the flower layer for spectral part a of PAR (400-525 nm)
TRANSB	auxiliary	TR_b	MJ MJ^{-1}	Transmission of PAR through the flower layer for spectral part b of PAR (252-700 nm)
TSEED	auxiliary		$^{\circ}\text{Cd}$	Base temperature for calculation of the temperature sum since 1/3 of the flowering period (at $\text{DVS}=2.333$) for calculation of normal seed weight
TSLAI	auxiliary		$^{\circ}\text{Cd}$	Temperature sum since the onset of crop regrowth in spring for calculation of LAI development ($\text{LAI} < 0.75$)

Continuation of Table II.1.

Acronym	Variable type	Symbol	Dimension	Description
TS0F	auxiliary		°Cd	Temperature sum since the onset of flowering for calculation of the critical period of seed set
TSSEED	auxiliary		°Cd	Temperature sum since 1/3 of the flowering period (at DVS=2.333) for calculation of normal seed weight
TWNOR	auxiliary		g m ⁻²	Total crop dry weight (including roots and dead leaves) (normal dry matter)
TWNM	output		g m ⁻²	Total crop dry weight (including roots and dead leaves) at maturity (normal dry matter); only for output
TWNTM	output		g m ⁻²	Total top crop dry weight (including dead leaves) at maturity (normal dry matter); only for output
VF	state	F_v		State of vernalization
VR	rate	dF_v/dt	d ⁻¹	Vernalization rate
YOM	auxiliary			Year of maturity; only for output

II-b: Symbols and acronyms

Table II.2: Symbols in LINTUL-BRASNAP.

Symbol	Acronym	Variable type	Dimension	Description
A_{if}	SPFL	parameter	m^2	Aea per individual flower
$a_{T,1}$	AT1	parameter	$d^{-1} \text{ } ^\circ C^{-1}$	Response of the development rate to temperature for the period from S till E
$a_{T,2}$	AT2	parameter	$d^{-1} \text{ } ^\circ C^{-1}$	Response of the development rate to temperature for the period from E till OF
$a_{T,3}$	AT3	parameter	$d^{-1} \text{ } ^\circ C^{-1}$	Response of the development rate to temperature for the period from OF till EF
$a_{T,4}$	AT4	parameter	$d^{-1} \text{ } ^\circ C^{-1}$	Response of the development rate to temperature for the period from EF till M
$dDVS/dt$	DVR	rate variable	d^{-1}	Daily development rate
D_{fp}	FPER	parameter	$^\circ C d$	Average duration of the flowering period
dF_v/dt	VR	rate	d^{-1}	Vernalization rate
D_{if}	DPFL	parameter	$^\circ C d$	Duration of flowering per flower
$dLAI/dt$	DLAI	rate	$m^2 m^{-2} d^{-1}$	Daily increment of the LAI
D_{sow}	SWD	parameter	d	Sowing day (Julian day)
DVS	DVS	state variable		State of development
dW_{res}/dt	DRECAR	rate variable	$g m^{-2}$	Daily increment of reserve carbohydrates (GCU)
dW_s/dt	DSEEDW	rate variable	$g m^{-2}$	Daily increment of seed weight (GCU)
dW_{tot}/dt	DTDW	rate variable	$g m^{-2}$	Daily increment of total crop dry weight (GCU)
F_1	PF1	parameter	m^{-2}	Parameter for calculation flower density
F_2	PF2	parameter		Parameter for calculation flower density
FAI	FAI	auxiliary	$m^2 m^{-2}$	Flower area index
$F_{I,r}$	FPAR	auxiliary	$MJ MJ^{-1}$	Fraction of incoming PAR at the bottom of the canopy
$F_{I,r,3.5}$	FPAR35	auxiliary	$MJ MJ^{-1}$	Fraction of incoming PAR at the bottom of the canopy when the development stage is 3.5 (near maturity)
$F_{oil,M}$	FOILM	parameter	$g g^{-1}$	Oil content of the seeds at maturity
$F_{oil,tt}$	ROILTB	function	$g g^{-1}$	Oil content of the seeds related to the temperature sum since 1/3 of the flowering period (since $DVS=2.333$)
F_p	DLF	auxiliary		Multiplication factor for the influence of photoperiod on the development rate
F_{pro}	FPROT	parameter	$g g^{-1}$	Protein content of the seeds during the seed filling phase
F_{res}	FRECAR	parameter	$g g^{-1}$	Fraction of reserve carbohydrates of total crop dry matter, just after flowering at $DVS=3.3$ (GCU)
F_v	VF	state		State of vernalization

Continuation of Table II.2.

Symbol	Acronym	Variable type	Dimension	Description
GRQ _{car}	GC	parameter	g g ⁻¹	Glucose units required for formation of carbohydrates
GRQ _{oil}	GO	parameter	g g ⁻¹	Glucose units required for formation of oil
GRQ _{pro}	GP	parameter	g g ⁻¹	Glucose units required for formation of protein
GRQ _s	GRQS	auxiliary	g g ⁻¹	Glucose units required for formation of seed dry matter
(GRQ _g) ⁻¹	CONVG	parameter	g g ⁻¹	Conversion factor of glucose units into green crop units dry matter
I ₀	PAR	forcing	MJ d ⁻¹	Daily incoming PAR (photosynthetic active radiation)
I _a	PARA	auxiliary	MJ d ⁻¹	Daily absorbed PAR (photosynthetic active radiation) by the green crop canopy
I _r	PARR	forcing	MJ d ⁻¹	Daily PAR at the bottom of the canopy (photosynthetic active radiation)
K _{LAI}	KLAI	parameter		Extinction coefficient for leaves
LAI	LAI	state variable	m m ⁻²	Leaf area index
LAI _s	SLAI	initial state	m ² m ⁻²	Initial leaf area index at the start of regrowth of the crop in early spring
LUE	LUE	auxiliary	g MJ ⁻¹	Light use efficiency
LUE _{DVS}	LUEDVS	parameter (i)	g MJ ⁻¹	Light use efficiency related to the development stage of the crop
N _f	NOFL	auxiliary	m ⁻²	Flower density
N _p	NPOD	auxiliary	m ⁻²	Pod density
N _s	NSEED	auxiliary	m ⁻²	Seed density
P	DAYLP	auxiliary	h	Photoperiodic daylength
P ₁	NP1	parameter	m ⁻²	Parameter for calculation pod density
P ₂	NP2	parameter		Parameter for calculation pod density
P _b	DLB	parameter	h	Basal photoperiod for development
P _{eff}	DLE	auxiliary	h	Effective photoperiod for development
P _i	FL	auxiliary	g g ⁻¹	Partitioning of daily crop growth to the leaves
P _{i,DVS}	FLDVS	function	g g ⁻¹	Partitioning of daily crop growth to the leaves, related to the development stage of the crop
P _s	FS	auxiliary	g g ⁻¹	Partitioning of daily crop growth to the seeds
P _{s,DVS}	FSDVS	function	g g ⁻¹	Partitioning of daily crop growth to the seeds, related to the development stage of the crop
P _{sat}	DLOPT	parameter	h	Saturating photoperiod
REF _{f,a}	GRA	auxiliary	MJ MJ ⁻¹	Reflection of incoming spectral part a of PAR (400-525 nm), by the flower layer
REF _{f,b}	GRB	auxiliary	MJ MJ ⁻¹	Reflection of incoming spectral part b of PAR (525-700 nm), by the flower layer
REF _g	REFGAF	parameter	MJ MJ ⁻¹	Reflection of PAR by the green canopy after the flowering period
REF _g	REFGBF	parameter	MJ MJ ⁻¹	Reflection of PAR by the green canopy before the flowering period
REF _g	REFLC	auxiliary	MJ MJ ⁻¹	Reflection of PAR by the green canopy

Continuation of Table II.2.

Symbol	Acronym	Variable type	Dimension	Description
$RF_{l,r,DVS}$	FPDVS	parameter		Response of the fraction of PAR at the bottom of the canopy near maturity ($DVS > 3.5$) to the increment of the development stage of the crop
RGR_{LAI}	RLAI	parameter	$m^2 m^{-2} ^\circ C d^{-1}$	Relative growth rate of the LAI with thermal time (if $LAI < 0.75$)
$RGR_{res,max}$	MDREC	parameter	$g g^{-1} d^{-1}$	Maximum depletion rate of the pool of reserve carbohydrates (GCU)
$R_{is,pot}$	PDMSW	auxiliary	$g ^\circ C d$	Daily potential individual seed growth per 1000. seeds (GCU)
$R_{is,pot,DVS}$	PDMSWT	function	$g ^\circ C d$	Daily potential individual seed growth per 1000. seeds (GCU), related to the development stage of the crop
$R_{v,max}$	MVR	parameter	d^{-1}	Maximum vernalization rate
S_1	NS1	parameter	m^{-2}	Parameter for calculation seed density
S_2	NS2	parameter		Parameter for calculation seed density
SLA	SLA	parameter	$m^2 g^{-1}$	Specific leaf area of young leaves
SPP_{pot}	PSPPP	parameter		Potential number of seeds per pod
$T_{b,1}$	TB1	parameter	$^\circ C$	Basic temperature for development in the period from S till E
$T_{b,2}$	TB2	parameter	$^\circ C$	Basic temperature for development in the period from E till OF
$T_{b,3}$	TB3	parameter	$^\circ C$	Basic temperature for development in the period from OF till EF
$T_{b,4}$	TB4	parameter	$^\circ C$	Basic temperature for development in the period from EF till M
T_{day}	TMG	auxiliary	$^\circ C$	Average daily temperature
T_{eff}	TME	auxiliary	$^\circ C$	The effective day temperature for crop development
TR	TRANFP	output	$MJ MJ^{-1}$	Transmission of PAR through the flower layer
TR	TRANS	auxiliary	$MJ MJ^{-1}$	Transmission of PAR through the flower layer
TR_a	TRANSA	auxiliary	$MJ MJ^{-1}$	Transmission of PAR through the flower layer for spectral part a of PAR (400-525 nm)
TR_b	TRANSB	auxiliary	$MJ MJ^{-1}$	Transmission of PAR through the flower layer for spectral part b of PAR (252-700 nm)
t_s	STIM	parameter	d	Time of start of regrowth of the crop in spring (julian day number plus 356. or plus 366. (leap-year)
tt	TS	auxiliary	$^\circ C d$	Temperature sum since the start of regrowth in spring for calculation of the LAI

Continuation of Table II.2.

Symbol	Acronym	Variable type	Dimension	Description
$T_{v,max}$	TMAX	parameter	°C	Maximum effective temperature for vernalization
$T_{v,min}$	TMIN	parameter	°C	Minimum effective temperature for vernalization
$T_{v,op,1}$	TOP1	parameter	°C	Lowest optimum temperature for vernalization
$T_{v,op,2}$	TOP2	parameter	°C	Highest optimum temperature for vernalization
W_{350}	TDW350	auxiliary	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at 350 °Cd after OF
W_{EF}	TDWEF	auxiliary	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at EF
W_{FF}	TDWFF	output	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at full flowering (=OF plus 7 d)
$W_{is,s}\ 10^{-3}$	IMSW	parameter	g	Initial individual seed weight (GCU per 1000 seeds) at the end of flowering (at DVS=3.0)
W_M	TDWM	output	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) at M
W_{OF}	TDWOF	auxiliary	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU) OF
W_{res}	RECAR	state variable	$g\ m^{-2}$	Weight of reserve carbohydrates (GCU)
W_s	SEEDW	state variable	$g\ m^{-2}$	Seed weight (GCU)
$W_{s,i}$	SMSW	auxiliary	$g\ m^{-2}$	Initial seed weight at the end of flowering
W_{top}	TDWTOP	output	$g\ m^{-2}$	Total top crop dry weight (including dead leaves and roots; GCU)
W_{tot}	TDW	state variable	$g\ m^{-2}$	Total crop dry weight (including dead leaves and roots; GCU)

Appendix III: Listing of LINTUL-BRASNAP 1.1

III-a: The model LINTUL-BRASNAP

```

*****
* LINTUL-BRASNAP 1.1
*
* Second version with some changes in the subroutine SUBNAP
* after the first test version for Cropsense, UK (BRASNAP1.for)
* Authors : B. Habekotte' (AB-DLO)
* Date : April 1995
* Controle : Daniel van Kraalingen (november, 1996)
* -----
*
* LINTUL-BRASNAP uses the FORTRAN Simulation Environment 2.1a
* With observation system
*
* FSE version 2.1 is described in:
* Kraalingen, D.W.G., van, 1995. The FSE system for crop simulation,
* version 2.1. Quantitative Approaches in System Analysis no. 1,
* AB-DLO/PE, Wageningen, 70 pp.
*
* Research Institute for Agrobiological and Soil Fertility (AB-DLO)
* Software Product Support
* P.O. Box 14,
* 6700 AA Wageningen, The Netherlands
* e-mail: sps@ab.agro.nl
*****

```

```

PROGRAM MAIN
CALL FSE
END

```

```

* -----
* SUBROUTINE MODELS
* Authors: Daniel van Kraalingen
* Date : 5-Jul-1993
* Purpose: This subroutine is the interface routine between the FSE-
* driver and the simulation models. This routine is called
* by the FSE-driver at each new task at each time step. It
* can be used by the user to specify calls to the different
* models that have to be simulated
*
* FORMAL PARAMETERS: (I=input,O=output,C=control,IN=init,T=time)
* name type meaning units class
* ----

```

```

* ITASK   I4   Task that subroutine should perform           -      I   *
* IUNITD  I4   Unit that can be used for input files         -      I   *
* IUNITO  I4   Unit used for output file                     -      I   *
* IUNITL  I4   Unit used for log file                         -      I   *
* FILEIT  C*   Name of input file with timer data           -      I   *
* FILEI1  C*   Name of input file no. 1                     -      I   *
* FILEI2  C*   Name of input file no. 2                     -      I   *
* FILEI3  C*   Name of input file no. 3                     -      I   *
* FILEI4  C*   Name of input file no. 4                     -      I   *
* FILEI5  C*   Name of input file no. 5                     -      I   *
* OUTPUT  L4   Flag to indicate if output should be done    -      I   *
* TERMNL  L4   Flag to indicate if simulation is to stop    -     I/O   *
* DOY     R4   Day number within year of simulation (REAL)   d      I   *
* IDOY    I4   Day number within year of simulation (INTEGER) d      I   *
* YEAR    R4   Year of simulation (REAL)                     y      I   *
* IYEAR   I4   Year of simulation (INTEGER)                  y      I   *
* TIME    R4   Time of simulation                             d      I   *
* STTIME  R4   Start time of simulation                      d      I   *
* FINTIM  R4   Finish time of simulation                     d      I   *
* DELT    R4   Time step of integration                      d      I   *
* LAT     R4   Latitude of site                               dec.degr. I   *
* WSTAT   C6   Status code from weather system               -      I   *
* WTRTER  L4   Flag whether weather can be used by model    -      O   *
* RDD     R4   Daily shortwave radiation                     J/m2/d   I   *
* TMMN    R4   Daily minimum temperature                     degrees C I   *
* TMMX    R4   Daily maximum temperature                     degrees C I   *
* VP      R4   Early morning vapour pressure                 kPa      I   *
* WN      R4   Average wind speed                             m/s      I   *
* RAIN    R4   Daily amount of rainfall                       mm/d     I   *
*
* Fatal error checks: none                                   *
* Warnings      : none                                       *
* Subprograms called: models as specified by the user       *
* File usage    : none                                       *
*-----*

```

```

SUBROUTINE MODELS (ITASK , IUNITD, IUNITO, IUNITL,
& FILEIT, FILEI1, FILEI2, FILEI3, FILEI4, FILEI5,
& OUTPUT, TERMNL,
& DOY , IDOY , YEAR , IYEAR ,
& TIME , STTIME, FINTIM, DELT ,
& LAT , LONG , ELEV , WSTAT , WTRTER,
& RDD , TMMN , TMMX , VP , WN, RAIN)
IMPLICIT REAL (A-Z)

```

```

* Formal parameters
INTEGER ITASK, IUNITD, IUNITO, IUNITL, IDOY, IYEAR
CHARACTER FILEIT*(*), FILEI1*(*), FILEI2*(*), FILEI3*(*),
CHARACTER FILEI4*(*), FILEI5*(*),
LOGICAL OUTPUT, TERMNL, WTRTER

```

CHARACTER WSTAT*6

* Local variables

* <none>

SAVE

* Only one model used here

CALL SUBNAP (ITASK, IUNITD, IUNITO, IUNITL,

& FILEI1, OUTPUT, TERMNL,

& DOY, YEAR,

& TIME, DELT,

& LAT, WSTAT, WTRTER,

& TMMN, TMMX, RDD)

RETURN

END

```

*-----*
*-----*
* Subroutine SUBNAP 1.1
*
* Second version with some changes in the subroutine SUBNAP
* after the first test version for Cropsense, UK (BRASNAP1.for)
* Authors : B. Habekotte' (AB-DLO)
* Date : April 1995
*
* SUBNAP calculates crop growth (since early spring) and
* phenological development (since sowing) of
* Winter oilseed rape (Brassica napus L.)
*
* SUBNAP 1.1 is described in:
* Habekotte, 1996. Description, parameterization and user guide of
* LINTUL-BRASNAP 1.1, Quantitative Approaches in Systems Analysis
* AB-DLO/PE, Wageningen, 66 pp (submitted).
*-----*
*
* changes in subroutine SUBNAP:
* > 21-07-1995: 'twntm= 0.95*twnm' instead of '=0.9*twnm'
*               = 'aboveground normal dry matter at final harvest'
*               (according to data of Habekotte and Smid (1992)
* > 01-12-1995: initial seed weight has a maximum of 25 g m-2
*               this is only necessary for explorations with changes
*               in seed density (numbers per m-2)
*-----*
*-----*

```

SUBROUTINE SUBNAP (ITASK, IUNITD, IUNITO, IUNITL,

& FILEI1, OUTPUT, TERMNL,

& DOY, YEAR,

& TIME, DELT,

```
&   LAT,WSTAT,WTRTER,
&   TMMN,TMMX,RDD)
```

```
IMPLICIT REAL (A-Z)
```

```
*-----Formal parameters
```

```
INTEGER   ITASK, IUNITD, IUNITO, IUNITL
LOGICAL   OUTPUT, TERMNL, WTRTER
CHARACTER*(*) FILEI1,WSTAT
```

```
*-----Standard local declarations
```

```
LOGICAL   INQOBS
INTEGER   L2, L6
PARAMETER (L2=25, L6=60)
INTEGER   IT1,IT3,IT4,IT11,IT12
INTEGER   IWVAR
REAL      ROILTB(L2)
REAL      FLDVS(L2), FSDVS(L2),LUEDVS(L2),PDMSWT(L6)
PARAMETER (TINY=1.E-4, TINY2=0.1)
CHARACTER WUSED*6
```

```
*-----The value of the fixed parameter 'fix' has to be equal
*   to the start day for simulation (stime: in timnap.dat),
*   a day in autumn before the day of sowing and after maturation;
*   the default value is day number 220.
PARAMETER (FIX = 220.)
```

```
SAVE
```

```
*   code for the use of RDD, TMMN, TMMX, VP, WN, RAIN (in that order)
*   a letter 'U' indicates that the variable is Used in calculations
DATA WUSED/'UUU---'/
```

```
*   Check weather data availability
```

```
IF (ITASK.EQ.1.OR.ITASK.EQ.2.OR.ITASK.EQ.4) THEN
  DO 10 IWVAR=1,6
```

```
*       is there an error in the IWVAR-th weather variable ?
```

```
IF (WUSED(IWVAR:IWVAR).EQ.'U' .AND.
```

```
&       WSTAT(IWVAR:IWVAR).EQ.'4') THEN
```

```
WTRTER = .TRUE.
```

```
TERMNL = .TRUE.
```

```
RETURN
```

```
END IF
```

```
10    CONTINUE
```

```
END IF
```

```
IF (ITASK.EQ.1) THEN
```

```
*-----*
*       Initialisation of states section       *
*       and reading parameters from input file *
*-----*
```

```

*-----send title to output file
*      CALL OUTCOM ('BRASNAP 1.1: winter oilseed rape growth
*                  and development')

*-----Initialization section
      IF (DELT.LT.1.0) CALL ERROR
      &      ('SUBNAP','DELT too small for SUBNAP')

*-----copy input file to output file once, open planap.dat:
      CALL RDINIT(IUNITD,IUNITL,FILEI1)

*-----reading parameter-values from planap.dat:

      CALL RDSREA ('SWD',SWD )
      CALL RDSREA ('STIM',STIM )
      CALL RDSREA ('TDW',TDW )
      CALL RDSREA ('SLAI',SLAI )

*-----other common parameters:
      CALL RDSREA ('VF' ,VF )
      CALL RDSREA ('TB1',TB1 )
      CALL RDSREA ('TB2',TB2 )
      CALL RDSREA ('TB3',TB3 )
      CALL RDSREA ('TB4',TB4 )
      CALL RDSREA ('AT1',AT1 )
      CALL RDSREA ('AT2',AT2 )
      CALL RDSREA ('AT3',AT3 )
      CALL RDSREA ('AT4',AT4 )
      CALL RDSREA ('TMIN',TMIN )
      CALL RDSREA ('TOP1',TOP1 )
      CALL RDSREA ('TOP2',TOP2 )
      CALL RDSREA ('TMAX',TMAX )
      CALL RDSREA ('MVR',MVR )
      CALL RDSREA ('DLB',DLB )
      CALL RDSREA ('DLOPT',DLOPT )

*-----reading parameters for growth and seed formation (common):
      CALL RDSREA ('CONVG',CONVG)
      CALL RDSREA ('FPROT',FPROT)
      CALL RDSREA ('FOILM',FOILM)
      CALL RDAREA ('ROILTB',ROILTB,L2, IT1)
      CALL RDSREA ('GO',GO)
      CALL RDSREA ('GP',GP)
      CALL RDSREA ('GC',GC)
      CALL RDAREA ('FLDVS',FLDVS,L2,IT3)
      CALL RDAREA ('FSDVS',FSDVS,L2,IT11)
      CALL RDAREA ('PDMSWT',PDMSWT,L6,IT12)
      CALL RDAREA ('LUEDVS',LUEDVS,L2,IT4)
      CALL RDSREA ('FPDVS',FPDVS)

```

```

CALL RDSREA ('KLAI',KLAI)
CALL RDSREA ('REFGBF',REFGBF)
CALL RDSREA ('REFGAF',REFGAF)
CALL RDSREA ('SPFL',SPFL)
CALL RDSREA ('DPFL',DPFL)
CALL RDSREA ('FPER',FPER)
CALL RDSREA ('PF1',PF1)
CALL RDSREA ('PF2',PF2)
CALL RDSREA ('RLAI',RLAI)
CALL RDSREA ('SLA',SLA)
CALL RDSREA ('PSPP',PSPP)
CALL RDSREA ('NP1',NP1)
CALL RDSREA ('NP2',NP2)
CALL RDSREA ('NS1',NS1)
CALL RDSREA ('NS2',NS2)
CALL RDSREA ('IMSW',IMSW)
CALL RDSREA ('FRECAR',FRECAR)
CALL RDSREA ('MDREC',MDREC)

```

```

CLOSE (IUNITD)

```

```

*-----Initialization of states

```

```

*-----for normal dry weight (g m-2):

```

```

HI2      = 0.
MSWN     = 0.
SWNOR    = 0.
SWNM     = 0.
TWNOR    = 0.
TWNM     = 0.

```

```

*-----for calculations in green crop units (GCU, g m-2)

```

```

* and other calculations:

```

```

DVSLUE   = 0.
DOF      = 0.
DFF      = 0.
D350     = 0.
DEF      = 0.
DOYM     = 0.
DVS      = 0.
DVSM     = 0.
DGRAF    = 1.
FOIL     = 0.
GRQST    = 0.
HI1      = 0.
LAI      = 0.
LAIM     = 0.
LUE      = 0.
MSWM     = 0.
NSEED    = 0.
NPOD     = 0.

```

```

NOFL      = 0.
NSNP      = 0.
RECAR     = 0.
RCALL     = 0.
RCALL1    = 0.
RCALL2    = 0.
RCALL4    = 0.
RCALL7    = 0.
RCALL8    = 0.
RCALL9    = 0.
RCALL10   = 0.
ROIL      = 0.
SEWM      = 0.
SEEDW     = 0.
SMSW      = 0.
TDWM      = 0.
TDWOF     = 0.
TDWFF     = 0.
TDW350    = 0.
TDWEF     = 0.
TDWTOP    = 0.
TSLAI     = 0.
TSOF      = 0.
TSSEED    = 0.
TLAI      = 0.
TOF       = 0.
TSEED     = 0.
WNTM      = 0.

*-----only for output:

      ELSE IF (ITASK.EQ.2) THEN
*-----*
*           Calculation of rates section           *
*-----*

*-----subroutine astro computes day-length from latitude and day-no:
      CALL ASTRO (LAT, DOY, DAYL, DAYLP)

*-----daily Temperature ( C): mean daily temprature :
      TMG = (TMMX + TMMN)*0.5

*-----daily effective temperature for calculation of temperature
*      sum with Tb=0.and with Tb=0.6870:
      TMM = LIMIT(0.,35.,TMG) - 0.
      TMS = LIMIT(0.6870,35.,TMG) - 0.6870

*-----temperatures with maximum phenological development rate:
      OPT1=TB1+30.
      OPT2=TB2+30.

```

OPT3=TB3+30.

OPT4=TB4+30.

*-----phenological development rate calculations:

```

IF (TIME.GE.SWD) THEN
  DLF = 0.
  DLE = 0.
  IF (DVS.GE.0..AND.DVS.LT.1.) THEN
    TME=LIMIT(TB1,OPT1,TMG) - TB1
    DVR=AMAX1(0.,AT1*TME)
  ELSE IF (DVS.LT.2..AND.DVS.GE.1.) THEN
    TME=LIMIT(TB2,OPT2,TMG) - TB2
    DLE=(LIMIT(DLB,DLOPT,DAYLP)) - DLB
    DLF=DLE*(1./(DLOPT-DLB))
    DVR=AMAX1(0.,DLF*VF*AT2*TME)
  ELSE IF (DVS.GE.2..AND.DVS.LT.3.) THEN
    TME=LIMIT(TB3,OPT3,TMG) - TB3
    DVR=AMAX1(0.,AT3*TME)
  ELSE IF (DVS.GE.3.) THEN
    TME=LIMIT(TB4,OPT4,TMG) - TB4
    DVR=AMAX1(0.,AT4*TME)
  ENDIF
ELSE
  DVR = 0.
  TME = 0.
  DLE = 0.
  DLF = 0.
ENDIF

```

*-----vernalization rates:

```

IF (DVS.GE.(1.0-TINY2).AND.DVS.LT.2..AND.VF.LT.1.) THEN
  IF ((TMG.LE.TOP2).AND.(TMG.GE.TOP1)) THEN
    VR=MVR
  ELSE IF ((TMG.GT.TMIN).AND.(TMG.LT.TOP1)) THEN
    VR=(TMG-TMIN)*(MVR/(TOP1-TMIN))
  ELSE IF ((TMG.GT.TOP2).AND.(TMG.LT.TMAX)) THEN
    VR = MVR - (TMG-TOP2)*(MVR/(TMAX-TOP2))
  ELSE IF ((TMG.GE.TMAX).OR.(TMG.LE.TMIN)) THEN
    VR=0.
  ENDIF
ELSE
  VR = 0.
ENDIF

```

*-----calculation of light absorbtion of the crop:

*-----a: calculation of incomming PAR:

```

*   RDD in J/m2/d en PAR in MJ/m2/d
    PAR = RDD*0.5*1.E-6

```

*-----b: calculation of availability of PAR for the growth of the crop

- * the transmission of PAR is lower than 1.0 during flowering because
- * of reflection and absorption of PAR by the yellow flowers.
- * (note: transfp is only calculated for output)

```

IF (TIME.GE.STIM) THEN
  FAI    = 0.
  GRA    = 0.
  GRB    = 0.
  TRANSA = 0.
  TRANSB = 0.
  IF (DVS.LT.2.) THEN
    TRANS = 1.
    REFLC = REFGBF
  ELSE IF (DVS.GE.2..AND.DVS.LE.3.) THEN
    FAI    = (DPFL/FPER)*NOFL*SPFL
    GRA    = 0.0183*(1.-EXP(-0.771*FAI))
    GRB    = 0.225*(1.-EXP(-0.506*FAI))
    TRANSA = 0.4*(1.-GRA)*EXP(-0.771*FAI)
    TRANSB = (1.-0.4)*(1.-GRB)*EXP(-0.506*FAI)
    TRANS  = TRANSA + TRANSB
    REFLC  = REFGBF
    IF (DVS.GE.2.5.AND.DVS.LT.2.6) THEN
      TRANFP = TRANS
    ENDIF
  ELSE IF (DVS.GE.3.) THEN
    TRANS = 1.
    REFLC = REFGAF
  ENDIF

  IF (DVS.LT.3.5) THEN
    FPAR35 = 0.
  ELSE IF (DVS.GE.3.5.AND.RCALL.EQ.0.) THEN
    FPAR35 = FPAR
    RCALL = 1.
  ENDIF

```

- *-----Near maturity PAR transmitted through the canopy increases
- * with development stage of the crop (fpdvs):

```

IF (DVS.LT.3.5) THEN
  FPAR = TRANS*(1.-REFLC)*EXP(-KLAI*LAI)
ELSE IF (DVS.GE.3.5.AND.RCALL.EQ.1.) THEN
  FPAR = FPAR35 + FPDVS*(DVS-3.5)
ENDIF

```

- *-----Transmitted PAR to the bottom of the canopy is assumed to be
- * minimal 5 % of incoming PAR:

```

IF (FPAR.LT.0.05) THEN
  FPAR = 0.05
ENDIF

```

- *----- (PAR at the bottom of the canopy):

```
PARR = FPAR*PAR
```

```
*-----d: absorbed PAR:
```

```
  PARA = PAR*TRANS*(1.-REFLC) - PARR
```

```
ELSE
```

```
  TRANS = 0.
```

```
  REFLC = 0.
```

```
  FAI   = 0.
```

```
  GRA   = 0.
```

```
  GRB   = 0.
```

```
  TRANSA = 0.
```

```
  TRANSB = 0.
```

```
  FPAR   = 0.
```

```
  FPAR35 = 0.
```

```
  PARR   = 0.
```

```
  PARA   = 0.
```

```
ENDIF
```

```
*-----calculation of daily growth of the crop in green crop units
```

```
*   via the lue (GCU; g/m2,day; before timst(iry) para=0. ):
*   (note: The global approach of crop growth related to absorbed light
*   by the green canopy in the model does not include influence
*   of temperature on crop growth).
```

```
*-----a: the lue related to the development stage of the crop:
```

```
  IF (TIME.GE.STIM.AND.DOY.LT.FIX) THEN
```

```
    LUEM = LINT (LUEDVS, IT4, DVS)
```

```
  ELSE
```

```
    LUEM = 0.
```

```
  ENDIF
```

```
*-----b: calculation of the lue from stim till the value at dvs=2.0:
```

```
  IF (TIME.LT.STIM.AND.DOY.GE.FIX) THEN
```

```
    DVSLUE = 0.
```

```
  ELSE IF (TIME.GE.STIM.AND.DOY.LT.FIX.AND.RCALL9.EQ.0.) THEN
```

```
    DVSLUE = DVS
```

```
    RCALL9 = 1.
```

```
  ENDIF
```

```
  IF (RCALL9.EQ.0..AND.DVS.LT.2.0) THEN
```

```
    ALFA = 0.
```

```
    LUE = 0.
```

```
  ELSE IF (RCALL9.EQ.1..AND.DVS.LT.2.0) THEN
```

```
    ALFA = (2.6818-1.145)/(2.0 - DVSLUE)
```

```
    LUE = 1.145 + ALFA*(DVS-DVSLUE)
```

```
*-----c: calculation of the lue after dvs=2.0:
```

```
  ELSE IF (DVS.GE.2.0) THEN
```

```
    ALFA = 0.
```

```
    LUE = LUEM
```

```

ENDIF

*-----d: calculation of the daily growth of the crop in GCU (g m-2):
IF (TIME.GE.STIM.AND.DOY.LT.FIX) THEN
    DTDW = LUE*PARA
ELSE
    DTDW = 0.
ENDIF

*-----calculation of daily increase of leaf area index (LAI)
*   a: calculation of daily increase in temperature sum (tslai)
IF (TIME.GE.STIM.AND.DOY.LT.FIX) THEN
    TLAI = TMM
ELSE
    TLAI = 0.
ENDIF
*   b: when LAI <= 0.75:
IF (TIME.GE.STIM.AND.LAI.LE.0.75.AND.DOY.LT.FIX) THEN
    DLAI = SLAI*RLAI*(EXP(RLAI*TSLAI))*TMM
    FL = 0.
*   c: when LAI>0.75:
ELSE IF (TIME.GE.STIM.AND.LAI.GT.0.75.AND.DOY.LT.FIX) THEN
    FL=LINT(FLDVS,IT3,DVS)
    DLAI=DTDW*FL*SLA
ELSE
    DLAI = 0.
    FL = 0.
ENDIF

*-----calculation of daily increase in temperature sum since
*   onset of flowering to determine the critical period of
*   seed set (350 Cd):
IF (DVS.GE.2.0) THEN
    TOF = TMM
ELSE
    TOF = 0.
ENDIF

*-----calculation of daily potential and actual seed growth
*   in green crop units (g/m2) with
*   the availability of reserves after the end of flowering.
*   the demand for reserves (in green crop units) is positive
*   or negative:

*   a) calculation of daily increase in temperature sum
*   since dvs = 2.333:
IF (DVS.GE.2.333) THEN
    TSEED = TMM
ELSE
    TSEED = 0.

```

ENDIF

*-----b) potential distribution of total dry matter to the seeds:
 FS=LINT(FSDVS,IT11,DVS)

*-----c) potential total seed growth:

```
IF (DVS.GE.3..AND.NSEED.GT.0.) THEN
  PDMSW=LINT(PDMSWT,IT12,DVS)
  PSEEDW = (PDMSW/1000.)*NSEED*TMS
  DEMAND = PSEEDW - DTDW*FS
ELSE
  PDMSW = 0.
  PSEEDW = 0.
  DEMAND = 0.
ENDIF
```

*-----d) total seed growth after dvs = 3.0 and since the number of seeds
 * is defined after the critical period of seedset;
 * with the use of reserve carbohydrates after DVS = 3.3:

```
IF (TIME.GE.STIM.AND.DVS.GE.3..AND.NSEED.GT.0.) THEN
  IF (DVS.LT.3.3.AND.DEMAND.GT.0.) THEN
    DSEEDW = DTDW*FS
  ELSE IF (DVS.LT.3.3.AND.DEMAND.LE.0.) THEN
    DSEEDW = PSEEDW
  ENDIF
  IF (DVS.LT.3.3) THEN
    DRECAR = 0.
  ENDIF
  IF (DVS.GE.3.3) THEN
    IF (DEMAND.LT.0.) THEN
      DRECAR = -DEMAND
      DSEEDW = PSEEDW
    ELSE IF (DEMAND.GE.0..AND.DEMAND.LE.(MDREC*RECAR)) THEN
      DSEEDW = PSEEDW
      DRECAR = -DEMAND
    ELSE IF (DEMAND.GE.0..AND.DEMAND.GT.(MDREC*RECAR)) THEN
      DSEEDW = DTDW*FS + MDREC*RECAR
      DRECAR = -MDREC*RECAR
    ELSE IF (DEMAND.GE.0..AND.RECAR.LE.0.) THEN
      DSEEDW = DTDW*FS
      DRECAR = 0.
    ENDIF
  ENDIF
ELSE
  DSEEDW = 0.
  DRECAR = 0.
ENDIF
```

*-----calculation of variable for making graphs with crop variables
 * versus number of days (dgr) since 1-1-year

```

IF(DOY.LT.FIX)THEN
  DGR   = 1.
ELSE
  DGR   = 0.
ENDIF

```

*-----green crop units balance check:

```

*   All crop parts are given in green crop units. Crop-parts that are
*   integrated are: total crop dry weight (tdw), seed weight (seedw)
*   and reserve-carbohydrates (recar);
*   after dvs = 3.3, total crop growth is distributed to the seeds with
*   a maximum of the potential growth rate of the seeds. When current
*   crop growth (dtdw) cannot cover the daily growth capacity of the
*   seeds, additional assimilates are drawn from the reserves up to a
*   maximum depletion rate. When current crop growth exceeds the growth
*   capacity of the seeds, the surplus is added to the reserve pool
*   The check-variable (gcudif) has to be equal to 0.0

```

```

IF(DVS.GE.3.3)THEN
  GCUIN = DSEEDW + DRECAR
  GCUFL = DTDW
  GCUDIF = (GCUIN - GCUFL)/NOTNUL(GCUIN)
ELSE
  GCUIN = 0.
  GCUFL = 0.
  GCUDIF = 0.
ENDIF

```

```

IF (OUTPUT) THEN

```

*-----output of states and rates (weights in g m-2)

```

  CALL OUTDAT (2,0, 'YEAR ' , YEAR)
  CALL OUTDAT (2,0, 'DOY'   , DOY)
  CALL OUTDAT (2,0, 'DVS'   , DVS )
  CALL OUTDAT (2,0, 'TDW'   , TDW)
  CALL OUTDAT (2,0, 'SEEDW' , SEEDW)

```

```

  IF (INQOBS (FILEI1,'TDW')) CALL OUTDAT
&      (2,0,'TDW_OBS',GETOBS (FILEI1,'TDW'))
  IF (INQOBS (FILEI1,'SEEDW')) CALL OUTDAT
&      (2,0,'SEEDW_OBS',GETOBS (FILEI1,'SEEDW'))
  IF (INQOBS (FILEI1,'LAI')) CALL OUTDAT
&      (2,0,'LAI_OBS',GETOBS (FILEI1,'LAI'))

```

```

ENDIF

```

*-----Determine the finish conditions of the simulation

```

IF (DVS.GE.4.1) TERMNL = .TRUE.

```

```
ELSE IF (ITASK.EQ.3) THEN
```

```
-----*
*           Calculation of states section           *
*-----*
```

```
*-----calculation of variable for making graphs with crop variables
* versus number of days since 1-1-year
  DGRAF = INTGRL(DGRAF,DGR,DELT)
```

```
*-----integration of the phenological development rate. Initial value
* of DVS in planap.dat and in rernap.dat
  DVS = INTGRL(DVS,DVR,DELT)
  IF (DVS.GE.(1.0-TINY).AND.RCAL10.EQ.0.) THEN
    DVS = 1.00
    RCAL10 = 1.
  ENDIF
```

```
*-----integration of vernalisation rate between dvs=1 and dvs=2 with
* a maximum of the vernalisation state of 1.
  VF = LIMIT(0.,1., INTGRL(VF,VR,DELT))
```

```
*-----calculation of the temperature sum since the
* onset of flowering (OF):
  TSOF = INTGRL(TSOF,TOF,DELT)
```

```
*-----calculation of the LAI (exponential growth till lai = 0.75)
* (a: calculation of the temperature sum (tb=0.) since initial
* growth day in spring (itim); b: calculation of the lai).
* It is assumed that the fraction of PAR absorbed by the green canopy
* after transmission through the flower layer, remains constant
* until near maturity. Therefore, the LAI is kept constant after
* the maximum value have been reached.
```

```
*-----calculation of the LAI:
* a: calculation of temperature sum since start of regrowth, lai
* development and integration of crop growth in spring at stim
  TSLAI = INTGRL(TS,TMM,DELT)
```

```
* b: initial condition of lai at stim in spring:
  IF (TIME.LE.STIM.AND.RCALL7.EQ.0.) THEN
    LAI = SLAI
    RCALL7 = 1.
  ENDIF
```

```
* c: calculation of the state of lai after stim in spring:
  LAI=INTGRL(LAI,DLAI,DELT)
```

```
*-----integration of daily growth of the crop in green crop units
```

```

*      (GCU; g/m2) since start of regrowth in spring at stim
      TDW = INTGRL(TDW,DTDW,DELT)

*-----calculation of total dry weight of the crop at the
*      onset of flowering (tdwof) and at the end of
*      flowering (tdwef) and the number of flowers
*      (related to tdwof), the number of pods (related to tdwef)
*      and the number of seeds (related to tdw350-tdwof)

      IF(DVS.GE.(2.-TINY).AND.RCALL1.EQ.0.)THEN
        DOF = DOY
        TDWOF = TDW
        RCALL1 = 1.
        NOFL = PF1 + PF2*TDWOF
      ENDIF
      IF(DOY.GE.(DOF+7.).AND.RCALL1.EQ.1..AND.DOY.LT.FIX)THEN
        TDWFF = TDW
        RCALL1 = 2.
      ENDIF
      IF(DVS.GE.(3.-TINY).AND.RCALL1.EQ.2.)THEN
        DEF = DOY
        TDWEF = TDW
        RCALL1=3.
        NPOD = NP1 + NP2*TDWEF
      ENDIF
*-----final characteristics and weights at maturity (GCU, g m-2):
      IF(DVS.GE.(4.-TINY).AND.RCALL1.EQ.3.)THEN
        DOYM = DOY
        DVSM = DVS
        LAIM = LAI
        TDWM = TDW
        MSWM = (SEEDW/NSEED)*1000.
        SEWM = SEEDW
        TDWTOP = 0.95*TDWM
        HI1 = SEWM/TDWTOP
        RCALL1 = 4.
      ENDIF

*-----calculation of seed density (m-2):
      IF(TSOF.GE.350..AND.RCALL8.EQ.0.)THEN
        D350 = DOY
        TDW350 = TDW
        NSEED = NS1 + NS2*(TDW350-TDWOF)
        RCALL8 = 1.
      ENDIF

      IF(RCALL8.EQ.1..AND.NPOD.GT.0.)THEN
        NSNP = NSEED/NPOD
        IF(NSNP.GT.PSPP)THEN
          NSEED = NPOD*PSPP
        
```

```
NSNP = NSEED/NPOD
```

```
RCALL8 = 2.0
```

```
ENDIF
```

```
ENDIF
```

```
*-----calculation of reserve-carbohydrates in green crop units :
```

```
* a: initial condition just after flowering at dvs = 3.3:
```

```
IF(DVS.GE.3.3.AND.RCALL4.EQ.0.)THEN
```

```
RECAR = FRECAR*TDW
```

```
RCALL4 = 1.
```

```
ENDIF
```

```
* b: the state of reserve-carbohydrates since dvs = 3.3:
```

```
RECAR = INTGRL(RECAR,DRECAR,DELT)
```

```
*-----calculation of total seed weight in green crop units (GCU g/m2)
```

```
*-----a: calculation of initial total seed weight at dvs >= 3.0:
```

```
* initial seed weight has a maximum of 25 g m-2::
```

```
IF(DVS.GE.3..AND.RCALL2.EQ.0..AND.NSEED.GT.0.)THEN
```

```
SEEDW = (IMSW/1000.)*NSEED
```

```
IF(SEEDW.GT.25.)THEN
```

```
SEEDW = 25.
```

```
ENDIF
```

```
RCALL2 = 1.
```

```
ENDIF
```

```
*-----b: total seed weight and individual seed weight (per 1000)
```

```
* calculated since the number of seeds are defined and since
```

```
* the end of flowering (dvs=3.0):
```

```
SEEDW = INTGRL(SEEDW,DSEEDW,DELT)
```

```
IF(SEEDW.GT.0.)THEN
```

```
SMSW = (SEEDW/NSEED)*1000.
```

```
ELSE
```

```
SMSW = 0.
```

```
ENDIF
```

```
*-----calculation of normal total dry weight and seed weight (g m-2):
```

```
* (weights of reserve carbohydrates remained in GCU)
```

```
*-----a: calculation of the temperature sum (tb=0.) since 1/3 flow. period
```

```
TSSEED = INTGRL(TSSEED,TSEED,DELT)
```

```
*-----b: calculation of normal seed weight in relation to oil,
```

```
* protein and carbohydrate content:
```

```
IF(SEEDW.GT.0.0)THEN
```

```
ROIL = LINT(ROILTB,IT1,TSSEED)
```

```
FOIL = ROIL*FOILM
```

```
GRQS = FOIL*GO + FPROT*GP + (1.-FOIL-FPROT)*GC
```

```
SWNOR = (SEEDW/CONVG)*(1./GRQS)
```

```
ENDIF
```

```
*-----c: calculation of normal total dry weight:
```

```

IF (SEEDW.GT.0.0) THEN
    TWNOR = TDW - SEEDW + SWNOR
ELSE
    TWNOR = TDW
ENDIF

*-----d: calculation of final normal dry weights at maturity (g m-2):
IF (DVS.GE.(4.-TINY).AND.RCALL1.EQ.4.) THEN
    TWNM = TWNOR
    SWNM = SWNOR
    MSWN = (SWNOR/NSEED)*1000.
    TWNTM = 0.95*TWNM
    HI2 = SWNM/TWNTM
    RCALL1 = 5.
ENDIF

ELSE IF (ITASK.EQ.4) THEN
*-----*
*                               Terminal section                               *
*-----*

*-----Define graph for output
*   use individual scale, small plot width for output

*   CALL OUTPLT (1, 'TDW')
*   CALL OUTPLT (1, 'LAI')
*   CALL OUTPLT (1, 'SEEDW')
*   CALL OUTPLT (6, 'Printplot of SUCROS')

*-----Store values of variables at maturity

*-----Calculate day of the year of full flowering (defined
*   as the day of onset of flowering plus seven days.
    DFF = DOF + 7.

    CALL OPSTOR ('YOM', YEAR)
    CALL OPSTOR ('DOF', DOF)
    CALL OPSTOR ('DEF', DEF)
    CALL OPSTOR ('DOYM', DOYM)
    CALL OPSTOR ('SEWM', SEWM)
    CALL OPSTOR ('TDWM', TDWM)
    CALL OPSTOR ('LAIM', LAIM)
    CALL OPSTOR ('NSEED', NSEED)
    CALL OPSTOR ('MSWM', MSWM)

ENDIF

RETURN
END

```

```

*-----*
* SUBROUTINE ASTRO
* Simplified version (B.Habekotte)
* Authors: Daniel van Kraalingen
* Date : 9-Aug-1987
* Modified by Jan Goudriaan 4 Febr 1988
* Modified by Jan Goudriaan and Kees Spitters 7 December 1989
* Purpose: This subroutine calculates astronomic daylength
*          and photoperiodic daylength
* FORMAL PARAMETERS: (I=input,O=output,C=control,IN=init,T=time)
* name      meaning
* ----      -
* DOY       Day number (Jan 1st = 1)
* LAT       Latitude of the site
* DAYL      Astronomical daylength (base = 0 degrees)
* DAYLP     Photoperiodic daylength (base = 0 degrees)
*
* FATAL ERROR CHECKS (execution terminated, message)
* condition: LAT > 67, LAT < -67
*
* SUBROUTINES and FUNCTIONS called : ERROR
* FILE usage : none
*-----*

```

```

SUBROUTINE ASTRO (LAT,DOY,DAYL,DAYLP)
IMPLICIT REAL (A-Z)
SAVE

```

```

*-----PI and conversion factor from degrees to radians
PARAMETER (PI=3.141592654, RAD=0.017453292)

```

```

*-----check on input range of parameters
IF (LAT.GT.67.) CALL ERROR ('ASTRO','LAT > 67')
IF (LAT.LT.-67.) CALL ERROR ('ASTRO','LAT < -67')

```

```

*-----declination of the sun as function of daynumber (DAY)
DEC = -ASIN (SIN (23.45*RAD)*COS (2.*PI*(DOY+10.)/365.))

```

```

*-----SINLD, COSLD and AOB are intermediate variables

```

```

SINLD = SIN (RAD*LAT)*SIN (DEC)
COSLD = COS (RAD*LAT)*COS (DEC)
AOB = SINLD/COSLD

```

```

*-----daylength (DAYL)
DAYL = 12.0*(1.+2.*ASIN (AOB)/PI)
DAYLP = 12.*(1.+2.*ASIN((-SIN(-4.*RAD)+SINLD)/COSLD)/PI)

```

```

RETURN
END

```

III-b: Input file PLANAP.DAT

```

*****
* PLANAP.DAT
* Input file for BRASN1_1.FOR
* Initial values and parameter values
* Parameter values are derived from experimental data of the
* variety JET NEUF. Parameterization is described in:
* Habekotte, 1996. Description, parameterization and listing of
* LINTUL-BRASNAP 1.1, Quantitative Approaches in Systems Analysis
* AB-DLO/PE, Wageningen, XX pp.
* For some parameters the original source of the experimental data
* are given
*****

*****
* initial conditions
*****

*---Day of sowing and startday for integration of development;
*   default value for the Netherlands is 237. (25-8)
SWD = 237.

*---Startday for integration of total dry matter in spring
*   default value for the Netherlands is 447. (22-3)
STIM = 447.

*---Initial value of the state of vernalization
VF = 0.

*---Initial value of leaf area index (lai, m2 m-2) in spring at
*   time = stim; calculated through (tdw*0.57*0.00981)
*   (Habekotte & Smid, 1992);
*   default value for the Netherlands is 1.12 m2 m-2

SLAI = 1.12

*---Initial value of total dry matter (g m-2) in spring
*   (de Boer & Langenhuysen, 1985);
*   default value for the Netherlands is 200. g m-2
TDW = 200.

*****
* Experimental data of total crop dry matter, seed dry matter and
* leaf area index during the growing season, that can be used for
* model evaluation; dummy values are filled in;
* see rernap.dat for an example.
*****

```

*---Total dry matter (g m⁻²) at harvest dates, including dead leaves:
TDW_OBS = -99.

*---Seed weight (g m⁻²) at harvest dates:
SEEDW_OBS = -99.

*---Leaf area index (m² m⁻²):
LAI_OBS = -99.

* COMMON DATA *

**** Parameters for phenological development of the crop *****
* (Habekotte, 1996a); Tx (C); ATx (d-1 C-1); MVR (d-1 C-1); DLx (h)

TB1 = 0.30236

TB2 = 0.54435

TB3 = 4.9163

TB4 = 0.6870

AT1 = 0.0077212325

AT2 = 0.00200831

AT3 = 0.0051036

AT4 = 0.0014651

TMIN = -3.71823

TOP1 = 0.72595

TOP2 = 5.37697

TMAX = 17.20219

MVR = 0.01455395

DLB = 5.7416

DLOPT = 14.8014

**** Parameters for crop growth *****

*---light use efficiency related to dvs (g MJ⁻¹) (Habekotte, 1996b)

* (1.145 is the value at stim in stead of at dvs=1.0, see model):

LUEDVS = -1.000, 0.0,
0.999, 0.0,
1.0 , 1.145,
2.0 , 2.6818,
3.5 , 1.9573,
4.2 , 0.0,
6.0 , 0.0

*---Increase of FPAR with development stage after dvs = 3.5

* based on: (d(fpar)/d(dvso), Habekotte & Smid, 1992)

FPDVS = 0.184

*---Extinction coefficient of light related to the leaf area index

* before flowering (Habekotte & Smid, 1992)

KLAI = 0.986

*---Reflection coefficient of the (green) canopy before flowering
 * (Yates & Steven, 1987; Leach, 1989)
 REFGBF = 0.046

*---Reflection coefficient of the (green) canopy after flowering
 * (Yates & Steven, 1987; Leach, 1987)
 REFGAF = 0.072

*---Flower area per flower (m²) (Habekotte, 1996)
 SPFL = 0.000213

*---Duration of flowering per flower (°Cd) (Habekotte, 1996)
 DPFL = 30.5 ! with base temperature: TB = 4.9163

*---Mean duration of the flowering period (Cd) (Habekotte, 1996a)
 FPER = 195.94 ! with base temperature TB = 4.9163

*---Parameters for calculation of the cumulative number of
 * flowers (1 m⁻²): linearly related to total dry
 * matter at the onset of flowering (Habekotte, 1993):
 PF1 = 6380.
 PF2 = 7.34

* distribution of dry matter to the leaves after the
 * initial phase of LAI development (LAI > 0.75)
 * (Habekotte & Smid, 1992; de Boer & Langenhuisen, 1985):
 FLDVS = 0.0 , 0.0,
 1.0 , 0.752,
 1.125, 0.752,
 1.425, 0.363,
 1.58, 0.330,
 1.78, 0.298,
 2.01, 0.265,
 2.321, 0.160,
 2.32101, 0.0,
 6.0 , 0.0

*---Relative increase rate of LAI (Cd⁻¹) when LAI < 0.75
 * (after Boer & Langenhuisen, 1985):
 RLAI = 0.007315

*---Specific leaf area of young leaves (m² g⁻¹)
 * (Habekotte & Smid, 1992)
 SLA = 0.0177

*---Fraction of total crop dry matter growth distributed to
 * the seeds as a function of the development stage
 * (Habekotte & Smid, 1992)

FSDVS = 0., 0.,
 2.41, 0.,
 2.81, 0.115,
 3.30, 1.0,
 5.0, 1.0

*---Initial mean seed weight (GCU g per 1000.) at the end of flowering
 * (at DVS=3.) (Habekotte & Smid, 1992)
 IMSW = 0.254

*---Potential individual seed growth per 1000 seeds (g Cd-1)
 * related to DVS since EF (Habekotte & Smid, 1992):
 PDMSWT = 0.000, 0.0, 2.999, 0.0,
 3.000, 0.0028382, 3.110, 0.006364,
 3.256, 0.015386, 3.330, 0.020086, 3.505, 0.020086,
 3.549, 0.017917,
 3.769, 0.004841, 4.355, 0.000045, 6.000, 0.0000

*---Factor for pdmsw (only used for sensitivity analysis):
 FPDMSW = 1.0

*---Fraction reserve carbohydrates of total dry matter (in GCU)
 * just after flowering (at dvs=3.3) (Habekotte & Smid, 1992)
 FRECAR = 0.049

*---Maximum relative depletion rate of the reserve pool in GCU (d-1)
 * (DRECAR = -MDREC*RECAR)
 * (Habekotte & Smid, 1992)
 MDREC = 0.0524

*---Parameters for calculation of the number of pods with seeds (m-2),
 * linearly related to total dry matter at the end of flowering:
 * (Habekotte, 1993)
 NP1 = -1021.
 NP2 = 6.76

*---Potential number of seeds per pod (Mendham et al., 1984):
 PSPP = 35.

*---Parameters for calculation of the number of seeds (m-2)
 * linearly related to dry matter production during the critical
 * period for seed set: a period from the onset of flowering (OF)
 * until 350 Cd since onset of flowering
 * (Habekotte, 1996b).
 NS1 = -7663.
 NS2 = 184.1

*---Conversion factor of glucose units dry matter into
 * green crop units of dry matter (=GRQG-1)
 * (Habekotte & Smid, 1992 and Penning de Vries et al., 1989):

CONVG = 0.671

*---Protein content of the seeds (circa constant)

* (Triboi-Blondel & Ribailier, 1988)

FPROT = 0.22

*---Oil-content of seeds at final harvest (average value of variety

* Jet Neuf from 1980-1989 in the flevopolders (RIJP-reports).

FOILM = 0.428

*---Conversion factors of glucose units required for seed dry matter:

*---oil (after Penning de Vries et al., 1989):

GO = 3.189

*---protein:

GP = 1.887

*---carbohydrates:

GC = 1.275

*---The relative change of the oil content of the seeds related to the

* maximum oil content at final harvest (foilm) and related

* to the temperature sum (tsseed) since the moment of seed initiation

* (estimated at 1/3 of the flowering period when DVS = 2.333

* (after Triboi-Blondel & Ribailier, 1988; Habekotte, 1996).

ROILTB = 0.0,0.00, 275.,0.051, 373.,0.41, 438.,0.69, 490.,0.82,
539.0,0.94, 588.,0.980, 641.,1.00, 1500.,1.00

III-c: Input file TIMNAP.DAT

```

*****
* TIMNAP.DAT
* Input file for BRASN1_1.FOR
* Habekotte, 1996. Description, parameterization and listing of
* LINTUL-BRASNAP 1.1, Quantitative Approaches in Systems Analysis
* AB-DLO/PE, Wageningen, XX pp.
*****
*****
* Name of weather directory:
*****
WTRDIR = 'C:\SYS\WEATHER\'
* WTRDIR = 'WEATHER_DATA:'

CNTR = 'NLD'      ! Nederland (NLD), Engeland (GBR)
ISTN = 1          ! Station number of weather data (1: Wageningen)
IYEAR = 1988      ! Year of weather data

*****
* Time variables and output file options
*****

STTIME = 220.     ! Start day of simulation: it has to be a day
                  ! in autumn before the sowing date (SWD) and after
                  ! the day of expected maturation; the fixed
                  ! parameter value 'fix' has to be equal to sttime
                  ! The default value is 220.

FINTIM = 620.     ! Finish time of simulation
PRDEL  = 20.      ! Time between consecutive outputs to file
DELT   = 1.       ! Time step of integration
IPFORM = 4        ! Format of output file (4 = normal table,
                  ! 5 = Tab-delimited (for Excel), 6=TTPLOT format)

DELTMP = 'N'      ! Switch variable what should be done with the
                  ! temporary output files (0 = do not delete,
                  ! 1 = delete)

COPINF = 'N'      ! Switch variable whether to copy the input files
                  ! to the output file ('N' = do not copy,
                  ! 'Y' = copy)

IFLAG  = 1101     ! Indicates where weather error and warnings
                  ! go (1101 means errors and warnings to log
                  ! file, errors to screen, see FSE manual)
                  ! The string array PRSEL contains the output
                  ! variables for which formatted tables have to be
                  ! made. One or more times there is a
                  ! series of variable names terminated by the word
                  ! <TABLE>.
                  ! The translator writes the variables in each

```

```
! PRINT statement to
* PRSEL = ! a separate table.
* 'DAYL ',
* 'RDPOT ',
* 'RDSUM ',
* 'RDACT ',
* ' <TABLE> '
```

```
**** List of output-day data for which output is requested
* 0. = no output data
```

```
IOBSD =
1989, 72,
1989, 87,
1989, 100,
1989, 114,
1989, 128,
1989, 142,
1989, 156,
1989, 170,
1989, 184,
1989, 191
```

III-d: Input file RERNAP.DAT

```

*****
* RERNAP.DAT
* Input file for BRASN1_1.FOR
* For excecution of reruns
* Habekotte, 1996. Description, parameterization and listing of
* LINTUL-BRASNAP 1.1, Quantitative Approaches in Systems Analysis
* AB-DLO/PE, Wageningen, 66 pp.
*****

**** Experimental data for model evaluation ****
* Wageningen, 1988-1989
* variety: Jet neuf
* exp1, first treatment: low plant density (lpd)
* Habekotte and Smid, 1992
*****

*****
* initial conditions
*****
*---Year, country, weather station, data number
CNTR  = 'NLD'      ! ,,
ISTN  = 1          ! ,,
IYEAR = 1988      ! see: timnap.dat

*---Date of sowing and startday for integration of development (23-8)
* (experiment 1, tr. 1, Habekotte and Smid, 1992):
SWD   = 236.

*---Startday for integration of total dry matter in (13-3) in spring
* (experiment 1, tr. 1, Habekotte and Smid, 1992):
STIM  = 438.

*---Initial value of leaf area index (lai) in spring at time = stim;
* calculated through (tdw*0.57*0.00981 (Habekotte & Smid, 1992):
* (experiment 1, tr. 1, Habekotte and Smid, 1992):
SLAI  = 0.787

*---Initial value of total dry matter (g/m2) in spring
* (experiment 1, tr. 1, Habekotte and Smid, 1992):
TDW   = 142.3

*****
* Experimental data of total crop dry matter during the growing
* season that can be used for evaluation of the model
*****
*---Total dry matter (g m-2, GCU) at harvest dates,

```

* including dead leaves

TDW_OBS =

1989., 72., 142.3,
 1989., 87., 172.1,
 1989., 100., 306.9,
 1989., 114., 417.6,
 1989., 128., 635.1,
 1989., 142., 915.8,
 1989., 156., 1224.7,
 1989., 170., 1513.1,
 1989., 184., 1561.6,
 1989., 191., 1485.3

**** Experimental data for calibration ****

* Wageningen, 1989-1990

* variety: Jet neuf

* exp2, first treatment: low seed rate (lsr)

* Habekotte and Smid, 1992

* initial conditions *

*---year, country, weather station, data number

CNTR = 'NLD'

ISTN = 1

IYEAR = 1989

SWD = 236.

STIM = 446.

SLAI = 3.32

TDW = 445.9

* Experimental data of total crop dry matter during the growing *

* season that can be used for evaluation of the model *

*---Total dry matter (g m⁻², GCU) at harvest dates,

* including dead leaves

TDW_OBS =

1990., 81., 445.9,
 1990., 108., 864.8,
 1990., 121., 1052.9,
 1990., 134., 1154.1,
 1990., 148., 1471.0,
 1990., 169., 1806.5,
 1990., 190., 1877.5

III-e: output file OP.OUT

RUNNUM	YOM	DOF	DEF	DOYM	SEWM	TDWM	LAIM	NSEED	MSWM
1	1989.0	104.	142.	187.	627.73	1654.5	3.1675	98425.	6.3777
2	1989.0	104.	142.	187.	626.31	1618.5	2.9543	98043.	6.3882
3	1990.0	101.	132.	184.	507.52	1815.7	5.3698	74135.	6.8460

III-f: output file BRASNAP.OUT

*-----

* Output table number : 0 (=first output table)

* Output table format : Table output

* Simulation results

TIME	YEAR	DOY	DVS	TDW	SEEDW
220.000	1988.0	220.00	.00000	200.00	.00000
240.000	1988.0	240.00	.33775	200.00	.00000
260.000	1988.0	260.00	1.0064	200.00	.00000
280.000	1988.0	280.00	1.0373	200.00	.00000
300.000	1988.0	300.00	1.0946	200.00	.00000
320.000	1988.0	320.00	1.1414	200.00	.00000
340.000	1988.0	340.00	1.1834	200.00	.00000
360.000	1988.0	360.00	1.2515	200.00	.00000
380.000	1989.0	14.000	1.3230	200.00	.00000
400.000	1989.0	34.000	1.3645	200.00	.00000
420.000	1989.0	54.000	1.4721	200.00	.00000
438.000	1989.0	72.000	1.6241	200.00	.00000
440.000	1989.0	74.000	1.6440	200.00	.00000
453.000	1989.0	87.000	1.7757	227.00	.00000
460.000	1989.0	94.000	1.8792	295.09	.00000
466.000	1989.0	100.00	1.9377	342.49	.00000
480.000	1989.0	114.00	2.0964	465.83	.00000
494.000	1989.0	128.00	2.3926	706.66	.00000
500.000	1989.0	134.00	2.5815	806.46	.00000
508.000	1989.0	142.00	3.0577	968.83	25.000
520.000	1989.0	154.00	3.2999	1226.8	183.67
522.000	1989.0	156.00	3.3246	1244.0	204.08
536.000	1989.0	170.00	3.6277	1497.1	486.95
540.000	1989.0	174.00	3.7322	1558.5	553.17
550.000	1989.0	184.00	3.9454	1635.3	616.79
556.000	1989.0	190.00	4.1187	1664.6	649.46

*-----

* Output from rerun set: 1

* Output table format : Table output

* Simulation results

TIME	YEAR	DOY	DVS	TDW	SEEDW	TDW_OBS
220.000	1988.0	220.00	.00000	142.30	.00000	-
240.000	1988.0	240.00	.44699	142.30	.00000	-

260.000	1988.0	260.00	1.0066	142.30	.00000	-
280.000	1988.0	280.00	1.0376	142.30	.00000	-
300.000	1988.0	300.00	1.0951	142.30	.00000	-
320.000	1988.0	320.00	1.1419	142.30	.00000	-
340.000	1988.0	340.00	1.1840	142.30	.00000	-
360.000	1988.0	360.00	1.2521	142.30	.00000	-
380.000	1989.0	14.000	1.3236	142.30	.00000	-
400.000	1989.0	34.000	1.3651	142.30	.00000	-
420.000	1989.0	54.000	1.4727	142.30	.00000	-
438.000	1989.0	72.000	1.6246	142.30	.00000	142.30
440.000	1989.0	74.000	1.6446	146.22	.00000	-
453.000	1989.0	87.000	1.7763	193.94	.00000	172.10
460.000	1989.0	94.000	1.8797	262.89	.00000	-
466.000	1989.0	100.00	1.9382	309.86	.00000	306.90
480.000	1989.0	114.00	2.0969	430.20	.00000	417.60
494.000	1989.0	128.00	2.3932	669.85	.00000	635.10
508.000	1989.0	142.00	3.0583	933.52	24.903	915.80
520.000	1989.0	154.00	3.3004	1191.3	183.52	-
522.000	1989.0	156.00	3.3251	1208.5	206.70	1224.7
536.000	1989.0	170.00	3.6282	1461.4	486.77	1513.1
540.000	1989.0	174.00	3.7328	1522.7	552.43	-
550.000	1989.0	184.00	3.9460	1599.3	615.43	1561.6
556.000	1989.0	190.00	4.1193	1628.6	647.86	-

*

* Output from rerun set: 2

* Output table format : Table output

* Simulation results

TIME	YEAR	DOY	DVS	TDW	SEEDW	TDW_OBS
220.000	1989.0	220.00	.00000	445.90	.00000	-
240.000	1989.0	240.00	.40606	445.90	.00000	-
260.000	1989.0	260.00	1.0117	445.90	.00000	-
280.000	1989.0	280.00	1.0396	445.90	.00000	-
300.000	1989.0	300.00	1.0978	445.90	.00000	-
320.000	1989.0	320.00	1.1492	445.90	.00000	-
340.000	1989.0	340.00	1.1698	445.90	.00000	-
360.000	1989.0	360.00	1.2354	445.90	.00000	-
380.000	1990.0	15.000	1.2643	445.90	.00000	-
400.000	1990.0	35.000	1.3741	445.90	.00000	-
420.000	1990.0	55.000	1.5159	445.90	.00000	-
440.000	1990.0	75.000	1.7029	445.90	.00000	-
460.000	1990.0	95.000	1.9599	579.63	.00000	-
480.000	1990.0	115.00	2.2879	879.06	.00000	-
500.000	1990.0	135.00	3.0736	1177.0	28.393	-
520.000	1990.0	155.00	3.4094	1564.0	248.66	-
540.000	1990.0	175.00	3.7912	1763.3	468.17	-
555.000	1990.0	190.00	4.1227	1828.3	525.13	1877.5

III-g: Example weather file

File: NLD1.990

```

*-----*
*   Country:   Netherlands
*   Station:   Wageningen
*   Year:      1990
*   Source:    Dep. of Meteorology, Wageningen Agricultural
*               University.
*   Author:    Willem Stol
* Longitude:   05 40 E
* Latitude:    51 58 N
* Elevation:   7 m.
* Comments:    Location Haarweg. Missing values are exchanged
*               with daily weather data from synoptic station
*               de Bilt, Netherlands Radiation on day 1 is
*               taken from NL7.990, old value was 8640 KJ/M**2/
*               D. Rainfall on day 328 is now 0 mm instead of -
*               0.2 mm.
*
* Columns:
* =====
* station number
* year
* day
* irradiation (kJ m-2 d-1)
* minimum temperature (degrees Celsius)
* maximum temperature (degrees Celsius)
* vapour pressure (kPa)
* mean wind speed (m s-1)
* precipitation (mm d-1)
*-----*
5.67  51.97      7. 0.00 0.00
-999 1990   1      3      1      1      1      1      1
    1 1990   1    770.  -0.2   0.7   0.820  2.8   0.0
-999 1990   2      1      1      1      1      1      1
    1 1990   2   1400. -0.4   2.2   0.600  1.3   0.0
    1 1990   3    620.  0.0   2.4   0.620  3.4   0.0
    1 1990   4   1250. -1.2   1.6   0.550  2.9   0.1
    1 1990   5    560.  1.6   3.7   0.690  2.2   0.0
etc....
    1 1990 360   1440.  2.3   8.8   0.760  7.2  11.7
    1 1990 361   1870.  3.2   8.4   0.680  6.3   1.9
    1 1990 362   2310.  0.9   8.1   0.730  5.4   0.0
    1 1990 363    710.  6.5  10.4   0.980  5.2   8.7
    1 1990 364   3230.  4.0   8.4   0.780  4.7   0.0
    1 1990 365   2380.  1.5   6.5   0.740  3.2   2.9

```

